

LLMs4OL: 大语言模型用于本体学习

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摘要。 我们提出了LLMs4OL方法, 该方法利用大语言模型 (LLMs) 进行本体学习 (OL)。大语言模型在自然语言处理方面取得了显著进展, 展示了它们在不同知识领域捕捉复杂语言模式的能力。我们的LLMs4OL范式研究了以下假设:
Can LLMs effectively apply their language pattern capturing capability to OL, which involves automatically extracting and structuring knowledge from natural language text? 为了验证这一假设, 我们采用零样本提示方法进行了全面评估。我们评估了九种不同的LLM模型系列, 用于三个主要的本体学习任务: 术语类型、分类法发现和非分类关系提取。此外, 评估还包括多种本体知识类型, 包括WordNet中的词汇语义知识、GeoNames中的地理知识以及UMLS中的医学知识。

所获得的实证结果表明, 基础大语言模型并不足以适用于需要高度推理能力和领域专业知识的本体构建。然而, 在有效微调后, 它们或许可以作为一种合适的助手, 缓解本体构建的知识获取瓶颈。

关键词: 大语言模型 · LLM · 本体 · 本体学习 · 提示 · 基于提示的学习。

1 引言

本体学习 (OL) 是人工智能 (AI) 和知识工程领域的一项重要研究方向, 因为它解决了在各个领域中知识获取和表示的挑战。OL涉及从文本信息中自动识别术语、类型、关系以及潜在的公理, 以构建本体 [29]。存在大量人类专家创建的本体实例, 从通用本体到特定领域本体, 例如: 统一医学语言系统 (UMLS) [8]、WordNet [40]、GeoNames [52]、Dublin Core元数据倡议 (DCMI) [64]、schema.org [19], 等。传统本体的创建依赖于领域专家的手动规范, 这可能耗时、昂贵、易出错, 并且在知识不断演变或领域专家不可用时不可行。因此, OL

LLMs4OL: Large Language Models for Ontology Learning

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Abstract. We propose the LLMs4OL approach, which utilizes Large Language Models (LLMs) for Ontology Learning (OL). LLMs have shown significant advancements in natural language processing, demonstrating their ability to capture complex language patterns in different knowledge domains. Our LLMs4OL paradigm investigates the following hypothesis: *Can LLMs effectively apply their language pattern capturing capability to OL, which involves automatically extracting and structuring knowledge from natural language text?* To test this hypothesis, we conduct a comprehensive evaluation using the zero-shot prompting method. We evaluate nine different LLM model families for three main OL tasks: term typing, taxonomy discovery, and extraction of non-taxonomic relations. Additionally, the evaluations encompass diverse genres of ontological knowledge, including lexicosemantic knowledge in WordNet, geographical knowledge in GeoNames, and medical knowledge in UMLS. The obtained empirical results show that foundational LLMs are not sufficiently suitable for ontology construction that entails a high degree of reasoning skills and domain expertise. Nevertheless, when effectively fine-tuned they just might work as suitable assistants, alleviating the knowledge acquisition bottleneck, for ontology construction.

Keywords: Large Language Models · LLMs · Ontologies · Ontology Learning · Prompting · Prompt-based Learning.

1 Introduction

Ontology Learning (OL) is an important field of research in artificial intelligence (AI) and knowledge engineering, as it addresses the challenge of knowledge acquisition and representation in a variety of domains. OL involves automatically identifying terms, types, relations, and potentially axioms from textual information to construct an ontology [29]. Numerous examples of human-expert created ontologies exist, ranging from general-purpose ontologies to domain-specific ones, e.g., Unified Medical Language System (UMLS) [8], WordNet [40], GeoNames [52], Dublin Core Metadata Initiative (DCMI) [64], schema.org [19], etc. Traditional ontology creation relies on manual specification by domain experts, which can be time-consuming, costly, error-prone, and impractical when knowledge constantly evolves or domain experts are unavailable. Consequently, OL

技术已经出现，用于从非结构化或半结构化来源（如文本文档和互联网）自动获取知识，并将其转换为结构化本体。对该领域的快速回顾表明，传统的OL方法基于词汇句法模式挖掘和聚类 [65,41,36,25,4,20,59,53,27,2,23,22]。相比之下，自然语言处理（NLP）通过大语言模型（LLMs） [45] 的最新进展为传统OL方法提供了一种有前景的替代方案。OL的最终目标是为知识获取和表示提供一种经济高效且可扩展的解决方案，从而在各个领域中实现更高效、更有效的决策。为此，我们介绍了LLMs4OL范式，并为其作为基础性第一步进行实证验证。

目前，还没有研究明确地针对本体学习训练大语言模型。因此，为了首次测试大语言模型在本体学习上的应用，我们进行了一些实验性考虑。首先是：大语言模型的特性是否足以支持本体学习？首先，大语言模型在广泛而多样的文本上进行训练，这与本体开发者需要具备广泛领域知识的需求相符。其次，大语言模型基于转换器的核心技术，通过促进其参数的快速扩展，实现了更高的语言建模复杂度。这些参数代表了词语之间的连接，使大语言模型能够理解非结构化文本，如句子或段落。此外，通过从词语连接中推断复杂的语言模式，大语言模型在各种任务中展现出类似人类的响应能力，这在“涌现式”人工智能领域得到观察。这种行为意味着执行明确训练范围之外的任务，例如生成可执行代码、不同体裁的文本和准确的文本摘要 [57,62]。大语言模型从简单的词语连接中推断模式、编码语言语义的能力，对于本体学习至关重要。本体通常依赖于从非结构化文本中分析和推断结构化信息连接，例如术语类型的分类体系和关系 [17]。因此，LLMs4OL假设——即大语言模型在本体学习中有益应用——在概念上具有合理性。

大语言模型正在快速发展中。在撰写本文时，据报道已有至少60种不同的大语言模型 [5]。这促使我们进行了第二个主要的实验考虑。针对*LLMs4OL*任务假设，应测试哪些大语言模型？对各种大语言模型的实证验证对于自然语言处理的发展以及为研究任务选择合适的模型至关重要。尽管在多种自然语言处理任务中表现出色，但大语言模型的有效性存在差异。为了LLMs4OL的基础工作，我们根据架构全面选择了八个不同模型家族，并在本文撰写时报告了当时的最佳性能。三个主要的大语言模型架构是编码器、解码器和编码器-解码器。用于验证的选定大语言模型包括：BERT [15] (仅编码器)；BLOOM [55],MetaAI的LLaMA [58], OpenAI的GPT-3 [9], GPT-3.5 [45], GPT-4 [46](全部为仅解码器)；以及BART [32] 和 Google的Flan-T5 [10] (编码器-解码器)。最近的研究表明，BERT在文本分类和命名实体识别 [15]，方面表现出色，BART在文本生成和摘要 [32],方面有效，而LLaMA在各种自然语言处理任务中，包括推理、问答和代码生成 [5]，表现出高精度。Flan-T5强调指令微调，并展现出强大的多任务性能。BLOOM独特的多语言方法在文本分类和序列标注 等任务中实现了稳健的性能。最后，GPT系列因其类人的文本生成能力而脱颖而出 [15]。在本工作中，我们首次旨在全面统一这些大语言模型，以评估它们在LLMs4OL范式下的有效性。

techniques have emerged to automatically acquire knowledge from unstructured or semi-structured sources, such as text documents and the web, and transform it into a structured ontology. A quick review of the field shows that traditional approaches to OL are based on lexico-syntactic pattern mining and clustering [65,41,36,25,4,20,59,53,27,2,23,22]. In contrast, recent advances in natural language processing (NLP) through Large Language Models (LLMs) [45] offer a promising alternative to traditional OL methods. The ultimate goal of OL is to provide a cost-effective and scalable solution for knowledge acquisition and representation, enabling more efficient and effective decision-making in a range of domains. To this end, we introduce the LLMs4OL paradigm and empirically ground it as a foundational first step.

Currently, there is no research explicitly training LLMs for OL. Thus to test LLMs for OL for the first time, we made some experimental considerations. The first being: *Do the characteristics of LLMs justify ontology learning?* First, LLMs are trained on extensive and diverse text, similar to domain-specific knowledge bases [50]. This aligns with the need for ontology developers to have extensive domain knowledge. Second, LLMs are built on the core technology of transformers that have enabled their higher language modeling complexity by facilitating the rapid scaling of their parameters. These parameters represent connections between words, enabling LLMs to comprehend the meaning of unstructured text like sentences or paragraphs. Further, by extrapolating complex linguistic patterns from word connections, LLMs exhibit human-like response capabilities across various tasks, as observed in the field of “emergent” AI. This behavior entails performing tasks beyond their explicit training, such as generating executable code, diverse genre text, and accurate text summaries [57,62]. Such ability of LLMs to extrapolate patterns from simple word connections, encoding language semantics, is crucial for OL. Ontologies often rely on analyzing and extrapolating structured information connections, such as term-type taxonomies and relations, from unstructured text [17]. Thus LLMs4OL hypothesis of LLMs’ fruitful application for OL appeared conceptually justified.

LLMs are being developed at a rapid pace. At the time of writing of this work, at least 60 different LLMs are reported [5]. This led to our second main experimental consideration. *Which LLMs to test for the LLMs4OL task hypothesis?* Empirical validation of various LLMs is crucial for NLP advancements and selecting suitable models for research tasks. Despite impressive performances in diverse NLP tasks, LLM effectiveness varies. For the foundational groundwork of LLMs4OL, we comprehensively selected eight diverse model families based on architecture and reported state-of-the-art performances at the time of this writing. The three main LLM architectures are encoder, decoder, and encoder-decoder. The selected LLMs for validation are: BERT [15] (encoder-only); BLOOM [55], MetaAI’s LLaMA [58], OpenAI’s GPT-3 [9], GPT-3.5 [45], GPT-4 [46] (all decoder-only); and BART [32] and Google’s Flan-T5 [10] (encoder-decoder). Recent studies show that BERT excels in text classification and named entity recognition [15], BART is effective in text generation and summarization [32], and LLaMA demonstrates high accuracy in various NLP tasks, including reason-

ing、问答和代码生成 [58]。Flan-T5强调指令微调，并展现出强大的多任务性能 [10]。BLOOM独特的多语言方法在文本分类和序列标注 [55]等任务中实现了稳健的性能。最后，GPT系列因其类人的文本生成能力而脱颖而出 [9,45,46]。在本工作中，我们首次旨在全面统一这些大语言模型，以评估它们在LLMs4OL范式下的有效性。

在两个实验考虑因素确定后，我们现在介绍LLMs4OL范式并强调我们的贡献。LLMs4OL围绕本体开发，包含以下基本元素 [38]: **1.** 一组描述术语词汇条目的字符串 L ，用于概念类型; **2.** 一组概念类型 T ; **3.** 一个类型层次结构分类法 H_T ; **4.** 一组由其域和范围限制描述的非分类法关系 R ，这些关系以关系异构结构排列 H_R ; 以及 **5.** 一组公理 A ，描述本体上的附加约束，使隐含事实显式化。本工作引入的LLMs4OL范式，将本体学习的三个核心方面作为任务，概述为以下研究问题（RQs）。

- **RQ1:** 术语类型任务 – 大语言模型在自动类型发现以构建本体方面的有效性如何? – **RQ2:** 类型分类法发现任务 – 大语言模型在识别类型分类法即类型之间的“是-属”层次结构方面的有效性如何? – **RQ3:** 类型非分类法关系抽取任务 – 大语言模型在发现类型之间非分类法关系方面的有效性如何?

这项工作的实证测试的多样性不仅在于所考虑的大语言模型，还在于所测试的本体知识领域。具体来说，我们测试了大语言模型在WordNet [40], 中的词汇语义知识、在GeoNames [1], 中的地理知识、在UMLS [7], 中的生物医学知识以及在schema.org [47]中的网页内容类型表示。为了对LLMs4OL进行实证验证，我们抓住机会包含了PubMedBERT [18], , 这是一个专门为生物医学领域设计的大语言模型，因此仅适用于UMLS。这一补充构成了先前引入的八个领域无关模型家族作为第九种模型类型。总而言之，我们主要的贡献是：

- LLMs4OL任务范式作为一个概念框架，用于利用大语言模型进行本体学习。 – LLMs4OL 概念的一个实现，利用定制的提示模板在三个特定任务的上下文中进行零样本本体学习，这些任务分别是术语类型、类型分类关系发现和类型非分类关系发现。这些任务在社区中众所周知的独特本体源上进行评估。我们的代码源，包括每个任务的模板和数据集，在此发布 <https://github.com/HamedBabaei/LLMs4OL>。 – 对八个最先进领域

独立的大语言模型类型（10个模型）以及一个第九个生物医学领域特定的大语言模型类型（第11个模型），以评估它们适用于本工作中考虑的各种本体学习任务。此外，最有效的大语言模型被微调，并且微调后的大语言模型结果针对我们的三个本体学习任务进行了报告。

ing, question answering, and code generation [58]. Flan-T5 emphasizes instruction tuning and exhibits strong multi-task performance [10]. BLOOM’s unique multilingual approach achieves robust performance in tasks like text classification and sequence tagging [55]. Lastly, the GPT series stands out for its human-like text generation abilities [9,45,46]. In this work, we aim to comprehensively unify these LLMs for their effectiveness under the LLMs4OL paradigm for the first time.

With the two experimental considerations in place, we now introduce the LLMs4OL paradigm and highlight our contributions. LLMs4OL is centered around the development of ontologies that comprise the following primitives [38]: **1.** a set of strings that describe terminological lexical entries L for conceptual types; **2.** a set of conceptual types T ; **3.** a taxonomy of types in a hierarchy H_T ; **4.** a set of non-taxonomic relations R described by their domain and range restrictions arranged in a heterarchy of relations H_R ; and **5.** a set of axioms A that describe additional constraints on the ontology and make implicit facts explicit. The LLMs4OL paradigm, introduced in this work, addresses three core aspects of OL as tasks, outlined as the following research questions (RQs).

- **RQ1:** *Term Typing Task* – How effective are LLMs for automated type discovery to construct an ontology?
- **RQ2:** *Type Taxonomy Discovery Task* – How effective are LLMs to recognize a type taxonomy i.e. the “is-a” hierarchy between types?
- **RQ3:** *Type Non-Taxonomic Relation Extraction Task* – How effective are LLMs to discover non-taxonomic relations between types?

The diversity of the empirical tests of this work are not only w.r.t. LLMs considered, but also the ontological knowledge domains tested for. Specifically, we test LLMs for lexico-semantic knowledge in WordNet [40], geographical knowledge in GeoNames [1], biomedical knowledge in UMLS [7], and web content type representations in schema.org [47]. For our empirical validation of LLMs4OL, we seize the opportunity to include PubMedBERT [18], a domain-specific LLM designed solely for the biomedical domain and thus applicable only to UMLS. This addition complements the eight domain-independent model families introduced earlier as a ninth model type. Summarily, our main contributions are:

- The LLMs4OL task paradigm as a conceptual framework for leveraging LLMs for OL.
- An implementation of the LLMs4OL concept leveraging tailored prompt templates for zero-shot OL in the context of three specific tasks, viz. term typing, type taxonomic relation discovery, and type non-taxonomic relation discovery. These tasks are evaluated across unique ontological sources well-known in the community. Our code source with templates and datasets per task are released here <https://github.com/HamedBabaei/LLMs4OL>.
- A thorough out-of-the-box empirical evaluation of eight state-of-the-art domain-independent LLM types (10 models) and a ninth biomedical domain-specific LLM type (11th model) for their suitability to the various OL tasks considered in this work. Furthermore, the most effective overall LLM is finetuned and subsequently finetuned LLM results are reported for our three OL tasks.

2 相关工作

存在三条相关的研究途径：从文本中进行本体学习、提示大语言模型以获取知识，以及大语言模型提示方法或提示工程。**从文本中进行本体学习。** 其中最早的方法之一 [22] 使用词汇句法模式从大量非结构化文本中提取新的词汇语义概念和关系，以增强 WordNet [40]。WordNet 是一个词汇数据库，包含概念（名词、动词等）的词汇本体和词汇语义关系（同义关系、下位关系等）。Hwang [23] 提出了一种构建特定应用领域动态本体的替代方法。该方法涉及使用代表高级领域类型的种子术语集，迭代地从非结构化文本中发现类型和分类层次结构。在每次迭代中，新发现的专门类型被纳入，并且算法检测语言特征之间的关系。该方法利用基于继承层次结构和集合运算的简单本体代数。Agirre 等人.[2] 通过从网络文档中提取主题相关的词语来增强 WordNet。这种独特的方法为 WordNet 添加了主题签名以丰富其内容。Kietz 等人.[27] 引入了 On-To-Knowledge 系统，该系统使用像 GermaNet [21] 或 WordNet 这样的通用核心本体作为基础结构。它旨在从企业内部网文本资源中发现特定领域的本体。对于概念提取和修剪，它采用统计术语频率计数启发式方法，而关联规则则应用于企业文本中关系识别。Roux 等人.[53] 提出了一种通过重用现有领域本体并通过从非结构化文本中提取的动词模式来增强遗传学本体的方法。他们的系统利用了语言工具，如词性标注器和句法解析器。Wagner[59] 通过对语料库进行统计分析，通过发现关系、向概念添加新术语，并通过自动获取动词偏好来获取概念，从而以非英语语言丰富 WordNet。Moldovan 和 Girju [42] 引入了文本知识获取 (KAT) 系统来增强 WordNet 的金融领域覆盖范围。他们的方法涉及四个阶段：(1) 从种子术语集中发现新概念，使用词典扩展概念列表；(2) 从新概念中识别词汇模式；(3) 从词汇模式中发现关系；(4) 使用知识分类算法将提取的信息集成到 WordNet 中。在 [4]，中提出了一种无监督方法，使用 NLP 技术如命名实体识别 (NER) 和词义消歧 (WSD) 来使用领域特定信息增强本体。该方法利用通用 NER 系统来揭示分类层次结构，并采用 WSD 通过查询互联网获取新术语并通过共现频率消歧来丰富现有同义词集。Khan 和 Luo [25] 使用聚类技术来发现新术语，利用 WordNet 进行分类。他们使用了受分子进化启发的自组织树算法 [16]，来建立本体层次结构。此外，Xu 等人. [65] 专注于通过基于 TFIDF 的单词语义分类器、基于已知关系和搭配的词汇句法模式查找器以及利用发现的词汇句法模式的关联规则提取器来自动获取特定领域的术语和关系。

2 Related Work

There are three avenues of related research: ontology learning from text, prompting LLMs for knowledge, and LLM prompting methods or prompt engineering. **Ontology Learning from Text.** One of the earliest approaches [22] used lexicosyntactic patterns to extract new lexicosemantic concepts and relations from large collections of unstructured text, enhancing WordNet [40]. WordNet is a lexical database comprising a lexical ontology of concepts (nouns, verbs, etc.) and lexico-semantic relations (synonymy, hyponymy, etc.). Hwang [23] proposed an alternative approach for constructing a dynamic ontology specific to an application domain. The method involved iteratively discovering types and taxonomy from unstructured text using a seed set of terms representing high-level domain types. In each iteration, newly discovered specialized types were incorporated, and the algorithm detected relations between linguistic features. The approach utilized a simple ontology algebra based on inheritance hierarchy and set operations. Agirre et al.[2] enhanced WordNet by extracting topically related words from web documents. This unique approach added topical signatures to enrich WordNet. Kietz et al.[27] introduced the On-To-Knowledge system, which utilized a generic core ontology like GermaNet [21] or WordNet as the foundational structure. It aimed to discover a domain-specific ontology from corporate intranet text resources. For concept extraction and pruning, it employed statistical term frequency count heuristics, while association rules were applied for relation identification in corporate texts. Roux et al.[53] proposed a method to expand a genetics ontology by reusing existing domain ontologies and enhancing concepts through verb patterns extracted from unstructured text. Their system utilized linguistic tools like part-of-speech taggers and syntactic parsers. Wagner [59] employed statistical analysis of corpora to enrich WordNet in non-English languages by discovering relations, adding new terms to concepts, and acquiring concepts through the automatic acquisition of verb preferences. Moldovan and Girju [42] introduced the Knowledge Acquisition from Text (KAT) system to enrich WordNet’s finance domain coverage. Their method involved four stages: (1) discovering new concepts from a seed set of terms, expanding the concept list using dictionaries; (2) identifying lexical patterns from new concepts; (3) discovering relations from lexical patterns; and (4) integrating extracted information into WordNet using a knowledge classification algorithm. In [4], an unsupervised method is presented to enhance ontologies with domain-specific information using NLP techniques such as NER and WSD. The method utilizes a general NER system to uncover a taxonomic hierarchy and employs WSD to enrich existing synsets by querying the internet for new terms and disambiguating them through cooccurrence frequency. Khan and Luo [25] employed clustering techniques to find new terms, utilizing WordNet for typing. They used the self-organizing tree algorithm [16], inspired by molecular evolution, to establish an ontology hierarchy. Additionally, Xu et al. [65] focused on automatically acquiring domain-specific terms and relations through a TFIDF-based single-word term classifier, a lexico-syntactic pattern finder based on known relations and collocations, and a relation extractor utilizing discovered lexico-syntactic patterns.

迄今为止，用于本体学习（OL）的突出方法主要基于词汇句法模式进行术语和关系抽取，以及聚类进行类型发现。否则，它们基于种子术语引导的启动方法。读者可参考关于此主题的更详细综述 [6,37]，以获得本体学习（OL）的全面方法论图景。传统自然语言处理（NLP）由分模块的管道定义，机器逐步获得语言、句法和语义层面的标注来处理文本。大语言模型（LLMs）为AI系统开启了一个新纪元，无需模块化NLP系统即可理解自然语言，而我们在本次工作中首次将这种自然语言用于本体学习（OL）任务。

提示大语言模型获取知识。大语言模型（LLMs）可以基于其知识处理和检索事实，这使得它们成为各种自然语言处理（NLP）任务的优秀零样本学习器。提示大语言模型（LLMs）意味着使用一个模板函数 $f_{prompt}(x)$ 作为输入，这是一个带有某些未填充槽位的文本提示输入，然后使用大语言模型（LLMs）以概率方式填充未填充的信息，以获得最终字符串 x' ，从中可以推导出最终输出 y [34]。LAMA：语言模型分析（Language Model Analysis） [51] 基准已被引入作为一项探测技术，用于分析单向语言模型（即Transformer-XL [12]）和双向语言模型（即BERT和ELMo [48]）中包含的事实和常识知识，使用知识三元组中的完形填空提示。他们展示了预训练语言模型（PLMs）在探测事实方面的潜力——其中事实被视为主谓宾三元组或问答对——通过将事实转换为完形填空模板，该模板用作语言模型（LM）的输入以填充缺失的标记。进一步的研究通过自动发现提示 [24]，微调大语言模型（LLMs）以更好地探测 [3,31,66]，或以纯粹的无监督方式从语言模型（LMs）中探测知识 [49]扩展了LAMA。这些研究分析了大语言模型（LLMs）编码各种语言和非语言事实的能力。这项分析仅限于预定义的事实，这些事实加强了大语言模型（LLMs）的传统语言知识，因此不能反映大语言模型（LLMs）如何学习概念。为解决此限制，Dalvi等人 [13]提出了一项提案，以探索和检验大语言模型（LLMs）学习的潜在概念，为BERT提供了新的视角。他们将概念定义为“一组有意义的词”，即可以根据词汇、形态等关系进行聚类。在另一项研究 [54]，中，他们提出了<样式id='19'>ConceptX</样式>框架，通过扩展他们对七个大语言模型（LLMs）在潜在空间分析中的研究，并将分组概念与人类定义的概念进行对齐。这些研究表明，使用大语言模型（LLMs）并访问概念潜在空间，使我们能够对概念进行分组，并将它们与预定义的类型和类型关系发现进行对齐。

提示工程。作为一种新兴学科，提示工程专注于为LLM设计最优指令，以实现成功任务执行。标准提示 [61]是指导LLM的基本方法。它允许用户制作自己的定制化“自我设计提示”，以有效与LLM [9] 交互，并提示它们直接用答案回应给定的提示指令。考虑手动制作的FLAN

Predominantly, the approaches for OL [60] that stand out so far are based on lexico-syntactic patterns for term and relation extraction as well as clustering for type discovery. Otherwise, they build on seed-term-based bootstrapping methods. The reader is referred to further detailed reviews [6,37] on this theme for a comprehensive overall methodological picture for OL. Traditional NLP was defined by modular pipelines by which machines were equipped step-wise with annotations at the linguistic, syntactic, and semantic levels to process text. LLMs have ushered in a new era of possibilities for AI systems that obviate the need for modular NLP systems to understand natural language which we tap into for the first time for the OL task in this work.

Prompting LLMs for Knowledge. LLMs can process and retrieve facts based on their knowledge which makes them good zero-shot learners for various NLP tasks. Prompting LLMs means feeding an input x using a *template function* $f_{prompt}(x)$, a textual string prompt input that has some unfilled slots, and then the LLMs are used to probabilistically fill the unfilled information to obtain a final string x' , from which the final output y can be derived [34]. The LAMA: LLanguage Model Analysis [51] benchmark has been introduced as a probing technique for analyzing the factual and commonsense knowledge contained in unidirectional LMs (i.e. Transformer-XL [12]) and bidirectional LMs (i.e. BERT and ELMo [48]) with cloze prompt templates from knowledge triples. They demonstrated the potential of pre-trained language models (PLMs) in probing facts – where facts are taken into account as subject-relation-object triples or question-answer pairs – with querying LLMs by converting facts into a cloze template which is used as an input for the LM to fill the missing token. Further studies extended LAMA by the automated discovery of prompts [24], finetuning LLMs for better probing [3,31,66], or a purely unsupervised way of probing knowledge from LMs [49]. These studies analyzed LLMs for their ability to encode various linguistic and non-linguistic facts. This analysis was limited to predefined facts that reinforce the traditional linguistic knowledge of the LLMs, and as a result do not reflect how concepts are learned by the LLMs. In response to this limitation, Dalvi et al. [13] put forward a proposal to explore and examine the latent concepts learned by LLMs, offering a fresh perspective on BERT. They defined concepts as “a group of words that are meaningful,” i.e. that can be clustered based on relations such as lexical, morphological, etc. In another study [54], they propose the framework *ConceptX* by extending their studies on seven LLMs in latent space analysis with the alignment of the grouped concepts to human-defined concepts. These works show that using LLMs and accessing the concept’s latent spaces, allows us to group concepts and align them to pre-defined types and type relations discovery.

Prompt Engineering. As a novel discipline, prompt engineering focuses on designing optimal instructions for LLMs to enable successful task performance. Standard prompting [61] represents a fundamental approach for instructing LLMs. It allows users to craft their own customized “self-designed prompts” to effectively interact with LLMs [9] and prompt them to respond to the given prompt instruction straightaway with an answer. Consider the manually crafted FLAN

集合 [35] 处理除OL之外的各种NLP任务作为示例。值得注意的是，某些问题的性质自然包含逐步思考的过程以得出答案。换句话说，待解决的问题可以分解为一系列前置中间步骤，然后才能到达最终解决方案。例如，算术或推理问题。为了可解释性，并提供语言模型在某种意义上“思考时间”帮助其更准确地回应，也有高级提示工程方法。首先，根据思维链（CoT） [63] 提示方法，提示指令被设计为指示LLM将复杂任务分解为一系列增量步骤，以引导解决方案。这有助于LLM逐步推理，并得出更准确和逻辑的结论。另一方面，为了需要探索或战略预览的任务，引入了思维树（ToT） [67]。ToT通过探索作为LLM一般问题解决中间步骤的思考，对CoT提示进行泛化。CoT和ToT都通过结合中间推理步骤和少样本或零样本 [28] 提示，解锁复杂推理能力。另一种解决更复杂任务的方法是使用分解提示[26]，，我们可以将LLM难以处理的任务进一步分解为更简单的可解决子任务，并将这些子任务委托给子任务特定的LLM。

鉴于本工作中引入的 LLMs4OL 任务范式，复杂的提示并非主要问题，因为我们的当前重点是初步探索任务以确定需要进一步改进的领域。我们希望在深入研究更复杂的 CoT、ToT 和分解式提示等技术之前，了解目前已取得的进展。一旦我们更清晰地了解模型在标准提示设置中的能力和局限性，我们就可以通过将 OL 视为逐步推理任务来考虑其他非标准提示工程技术。

3 LLMs4OL 任务范式

本体学习大语言模型（LLMs4OL）任务范式提供了一个概念框架，旨在加速由领域专家独家进行、耗时且昂贵的本体构建，使其达到一个与强大的人工智能方法（如大语言模型）相结合的水平，以获得高质量的本体学习结果；理想情况下，该范式仅涉及领域专家在验证周期中的参与。理论上，通过正确的公式化表述，所有与本体学习相关联的任务都适合LLMs4OL任务范式。本体学习任务基于本体原语 [38]，，包括词汇条目 L 、概念类型 T 、类型的层次分类体系 H_T 、异构结构中的非分类关系 R ，以及一套用于描述本体约束和推理规则的公理 A 。为了处理这些原语，本体学习任务 [44] 包括：1) 语料库准备——为构建本体选择和收集源文本。2) 术语提取——识别和提取相关术语。3) 术语类型——将相似术语分组为概念类型。4) 分类体系构建——在类型之间建立“是”关系层次。5) 关系提取——识别“是”关系之外的语义关系。

collection [35] addressing diverse NLP tasks other than OL as an exemplar. Notably, the nature of some problems naturally encompass a step-by-step thought process for arriving at the answer. In other words, the problem to be solved can be decomposed as a series of preceding intermediate steps before arriving at the final solution. E.g., arithmetic or reasoning problems. Toward explainability and providing language models in a sense “time to think” helping it respond more accurately, there are advanced prompt engineering methods as well. As a first, as per the Chain-of-Thought (CoT) [63] prompting method, the prompt instruction is so crafted that the LLM is instructed to break down complex tasks as a series of incremental steps leading to the solution. This helps the LLM to reason step-by-step and arrive at a more accurate and logical conclusion. On the other hand Tree-of-Thoughts (ToT) [67] has been introduced for tasks that require exploration or strategic lookahead. ToT generalizes over CoT prompting by exploring thoughts that serve as intermediate steps for general problem-solving with LLMs. Both CoT and ToT unlock complex reasoning capabilities through intermediate reasoning steps in combination with few-shot or zero-shot [28] prompting. Another approach for solving more complex tasks is using decomposed prompting [26], where we can further decompose tasks that are hard for LLMs into simpler solvable sub-tasks and delegate these to sub-task-specific LLMs.

Given the LLMs4OL task paradigm introduced in this work, complex prompting is not a primary concern, as our current focus is on the initial exploration of the task to identify the areas where we need further improvement. We want to understand how much we have accomplished so far before delving into more complex techniques like CoT, ToT, and decomposed prompting. Once we have a clearer picture of the model’s capabilities and limitations in a standard prompting setting, we can then consider other than standard prompt engineering approaches by formulating OL as a stepwise reasoning task.

3 The LLMs4OL Task Paradigm

The Large Language Models for Ontology Learning (LLMs4OL) task paradigm offers a conceptual framework to accelerate the time-consuming and expensive construction of ontologies exclusively by domain experts to a level playing field involving powerful AI methods such as LLMs for high-quality OL results; consequently and ideally involving domains experts only in validation cycles. In theory, with the right formulations, all tasks pertinent to OL fit within the LLMs4OL task paradigm. OL tasks are based on ontology primitives [38], including lexical entries L , conceptual types T , a hierarchical taxonomy of types H_T , non-taxonomic relations R in a heterarchy H_R , and a set of axioms A to describe the ontology’s constraints and inference rules. To address these primitives, OL tasks [44] include: 1) Corpus preparation - selecting and collecting source texts for ontology building. 2) Terminology extraction - identifying and extracting relevant terms. 3) Term typing - grouping similar terms into conceptual types. 4) Taxonomy construction - establishing ”is-a” hierarchies between types. 5) Relationship extraction - identifying semantic relationships beyond ”is-

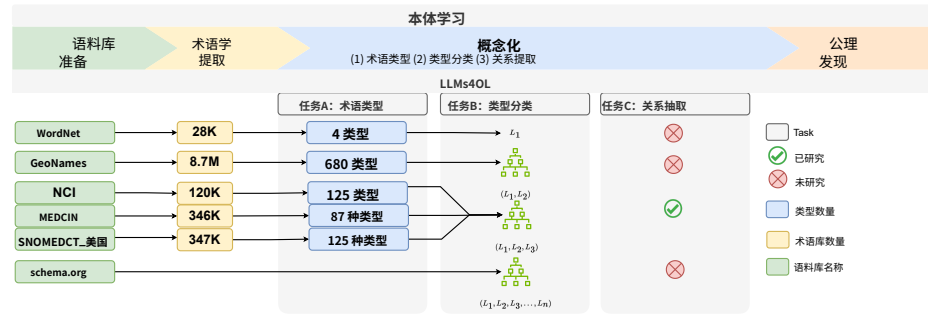


图1。 LLMs4OL任务范式是一个用于各种知识领域本体学习的端到端框架，即词汇语义学（WordNet）、地理学（GeoNames）、生物医学（NCI、MEDICIN、SNOMEDCT）和网页内容类型（schema.org）。这项工作中经验验证的三个本体学习任务在蓝色箭头内描绘，与更大的LLMs4OL范式对齐。

6) 公理发现——为本体寻找约束和推理规则。这六项任务构成了LLMs4OL任务范式。参见图1，了解所提出的LLMs4OL概念框架。

在本工作中，我们使用大语言模型（LLMs）作为未来研究的基础，通过实证方法验证了三种核心本体学习（OL）任务。然而，传统人工智能范式仅依赖于在明确训练的任务上测试模型，而这对大语言模型并不适用。相反，我们将大语言模型用于本体学习的表现视为一种“涌现”行为 [57,62]，即它们在没有明确训练的情况下，仍能在广泛任务上生成响应。解开大语言模型涌现能力的关键在于通过精心设计的提示来提示它们的知识，正如GPT-3 [9]，所推广的那样。如前所述（参见第2节），针对大语言模型的提示工程是一项新兴的人工智能子学科。在此过程中，预训练语言模型接收一个提示（例如自然语言陈述），以生成响应，而无需进一步训练或对其参数进行梯度更新 [34]。根据底层大语言模型预训练目标的不同，提示可分为两种主要类型：完形填空提示 [50,11]，它涉及在未完成的句子或段落中填写空白，适用于掩码语言模型预训练；以及前缀提示 [33,30]，它根据给定的起始短语生成文本，并为底层模型提供更多设计适应性。先前介绍的大语言模型本体学习（LLMs4OL）范式已通过为每个任务和模型设计相应的提示函数 $f_{prompt}(\cdot)$ 对三种选定的本体学习任务进行了实证验证。

任务A - 术语类型。 为某个词汇项发现了一种通用类型。

通用的完形填空提示模板是 $f_{c-prompt}^A(L) := [S?]. [L] [P_{domain}] is a [MASK].$ ，其中 S 是可选的上下文句子， L 是需要填空的词汇项， P_{domain} 是领域规范，而特殊 $MASK$ 标记是模型预期输出的类型。由于提示设计是决定大语言模型如何响应的重要因素，因此利用了该通用模板的八种不同提示模板实例，并报告了最终结果

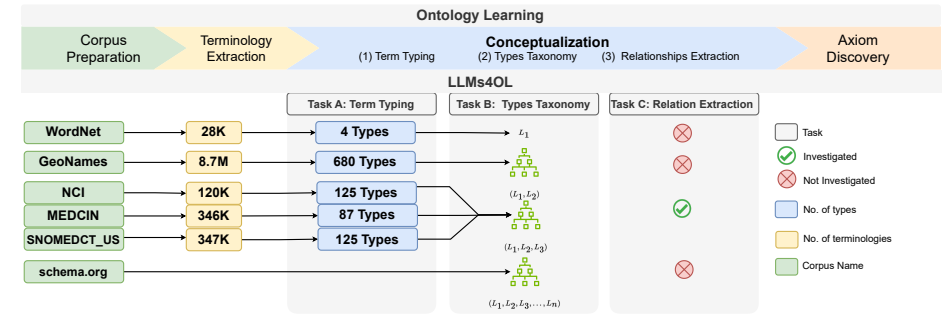


Fig. 1. The LLMs4OL task paradigm is an end-to-end framework for ontology learning in various knowledge domains, i.e. lexicosemantics (WordNet), geography (GeoNames), biomedicine (NCI, MEDICIN, SNOMEDCT), and web content types (schema.org). The three OL tasks empirically validated in this work are depicted within the blue arrow, aligned with the greater LLMs4OL paradigm.

a.” 6) Axiom discovery - finding constraints and inference rules for the ontology. This set of six tasks forms the LLMs4OL task paradigm. See Figure 1 for the proposed LLMs4OL conceptual framework.

In this work, we empirically ground three core OL tasks using LLMs as a foundational basis for future research. However, traditional AI paradigms rely on testing models only on explicitly trained tasks, which is not the case for LLMs. Instead, we test LLMs for OL as an ”emergent” behavior [57,62], where they demonstrate the capacity to generate responses on a wide range of tasks despite lacking explicit training. The key to unraveling the emergent abilities of LLMs is to prompt them for their knowledge, as popularized by GPT-3 [9], via carefully designed prompts. As discussed earlier (see section 2), prompt engineering for LLMs is a new AI sub-discipline. In this process, a pre-trained language model receives a prompt, such as a natural language statement, to generate responses without further training or gradient updates to its parameters [34]. Prompts can be designed in two main types based on the underlying LLM pretraining objective: cloze prompts [50,11], which involve filling in blanks in an incomplete sentence or passage and suit masked language modeling pre-training; and prefix prompts [33,30], which generate text following a given starting phrase and offer more design adaptability to the underlying model. The earlier introduced LLMs4OL paradigm is empirically validated for three select OL tasks using respective prompt functions $f_{prompt}(\cdot)$ suited to each task and model.

Task A - *Term Typing*. A generalized type is discovered for a lexical term.

The generic cloze prompt template is $f_{c-prompt}^A(L) := [S?]. [L] [P_{domain}] is a [MASK].$ where S is an optional context sentence, L is the lexical term prompted for, P_{domain} is a domain specification, and the special $MASK$ token is the type output expected from the model. Since prompt design is an important factor that determines how the LLM responds, eight different prompt template instantiations of the generic template were leveraged with final results reported

以寻找最佳模板。例如，如果 WordNet 是基础本体，则提示的词汇项词性类型。在这种情况下，模板 1 是 “[S]. [L] 词性是一个 [MASK]。” 请注意这里的 “[P_{domain}]” 是词性。模板 2 是 “[S]. [L] 词性是一个 [MASK]。” 请注意这里的 “[P_{domain}]” 是“词性”。类似地，从通用模板创建了八种不同的提示变体。然而， “[P_{domain}]” 的规范取决于本体的知识领域

前缀提示模板重用了完形填空提示模板，但附加了一条额外的“指令”句子，并将特殊的 [MASK] token 替换为空白或“?” 符号。泛化地说，它是 $f_{p-prompt}^A(T) = [instruction] + f_{c-prompt}^A(T)$ ，其中指令是“对以下句子执行句子补全：”。基于从通用完形填空模板提示中创建的八个变体，随后为 LLMs 的前缀提示创建了八个模板变体，并报告了最佳模板结果。

任务B - 分类法发现。这里发现了一对类型之间的分类层次结构。

通用完形填空提示模板是 $f_{c-prompt}^B(a, b) := [a|b] \text{ is } [P_{hierarchy}] \text{ of } [b|a]. \text{ This statement is } [MASK]$. 其中 (a, b) 或 (b, a) 是类型对， $P_{hierarchy}$ 表示如果模板初始化用于自顶向下的分类法发现，则指示超类关系；否则，如果模板初始化用于自底向上的分类法发现，则指示子类关系。在任务B中，对于给定的类型对，特殊 [MASK] token 的预期模型输出为真或假。

与术语类型类似，为通用模板创建了八个模板变体。其中四个基于自顶向下的分类法发现。例如，“[a] 是 [b] 的超类。该陈述是 [MASK]。” 请注意，此处 $[P_{hierarchy}]$ 是“超类”。其他三个模板基于 $[P_{hierarchy}] \in$ 父类、超类型、祖先类。另外四个基于自底向上的分类法发现的模板实例化基于 $[P_{hierarchy}] \in$ 子类、子类、子类型、后代类。因此，针对适用的 LLM，每个模板实例化运行了八个实验，并报告了最佳模板的结果。

前缀提示模板同样重用了完形填空提示模板，将 [MASK] token 替换为空白或“?” 符号。它是 $f_{p-prompt}^B(a, b) = [instruction] + f_{c-prompt}^B(a, b)$ ，指令为“识别以下陈述是否为真或假：”

任务C - 非分类法关系抽取。该任务发现类型之间的非分类法语义异层级关系。

完形填空提示模板是 $f_{c-prompt}^C(h, r, t) := [h] \text{ is } [r] [t]. \text{ This statement is } [MASK]$. 其中 h 是一个头类型， t 是一个尾类型，而 r 是 h 和 r 之间的非分类关系。为了支持发现一个可以由 1-M 关系基数组成的异构结构，对于给定的关系，本体中所有可能的类型对都被创建了。对于 [MASK] 标记的预期输出再次是 true 或 false。注意，与任务A和B不同，给定的模板被原样使用，没有创建它的变体。

for the best template. E.g., if WordNet is the base ontology, the part-of-speech type for the lexical term is prompted. In this case, template 1 is “[S]. [L] POS is a [MASK].” Note here “[P_{domain}]” is POS. Template 2 is “[S]. [L] part of speech is a [MASK].” Note here “[P_{domain}]” is “part of speech.” In a similar manner, eight different prompt variants from the generic template were created. However, the specification of “[P_{domain}]” depended on the ontology’s knowledge domain.

The prefix prompt template reuses the cloze prompt template but appends an additional “instruction” sentence and replaces the special [MASK] token with a blank or a “?” symbol. Generically, it is $f_{p-prompt}^A(T) = [instruction] + f_{c-prompt}^A(T)$, where the instruction is “Perform a sentence completion on the following sentence:” Based on the eight variations created from the generic cloze template prompt, subsequently eight template variations were created for the prefix prompting of the LLMs as well with best template results reported.

Task B - *Taxonomy Discovery*. Here a taxonomic hierarchy between pairs of types is discovered.

The generic cloze prompt template is $f_{c-prompt}^B(a, b) := [a|b] \text{ is } [P_{hierarchy}] \text{ of } [b|a]. \text{ This statement is } [MASK]$. Where (a, b) or (b, a) are type pairs, $P_{hierarchy}$ indicates superclass relations if the template is initialized for top-down taxonomy discovery, otherwise indicates subclass relations if the template is initialized for bottom-up taxonomy discovery. In Task B, the expected model output for the special [MASK] token for a given type pair was true or false.

Similar to term typing, eight template variations of the generic template were created. Four of which were predicated on the top-down taxonomy discovery. E.g., “[a] is the superclass of [b]. This statement is [MASK].” Note here, $[P_{hierarchy}]$ is “superclass”. Other three templates were based on $[P_{hierarchy}] \in$ parent class, supertype, ancestor class. And four more template instantiations predicated on the bottom-up taxonomy discovery were based on $[P_{hierarchy}] \in$ subclass, child class, subtype, descendant class. Thus eight experiments per template instantiation for the applicable LLM were run and the results from the best template were reported.

The prefix prompt template, similarly, reuses the cloze prompt template with the [MASK] token replaced with a blank or “?” symbol. It is $f_{p-prompt}^B(a, b) = [instruction] + f_{c-prompt}^B(a, b)$, with instruction “Identify whether the following statement is true or false:”

Task C - *Non-Taxonomic Relation Extraction*. This task discovers non-taxonomic semantic heterarchical relations between types.

The cloze prompt template is $f_{c-prompt}^C(h, r, t) := [h] \text{ is } [r] [t]. \text{ This statement is } [MASK]$. Where h is a head type, t is a tail type, and r is a non-taxonomic relationship between h and r . To support the discovery of a heterarchy that can consist of a 1-M relational cardinality, for a given relation, all possible type pairs of the ontology were created. The expected output for the [MASK] token was again true or false. Note, unlike in Task A and B, the given template was used as is and no variations of it were created.

再次，前缀提示模板重用了完形填空提示模板作为其他任务，其指令与任务B类似。它是 $f_{p-prompt}^C(h, r, t) = [instruction] + f_{c-prompt}^C(h, r, t)$

4 LLMs4OL - 三本体学习任务评估

4.1 评估数据集 - 本体知识源

为全面评估前文所述的三个本体学习任务，我们涵盖了多种本体知识领域来源。通常，在所有任务中，有四个知识领域被代表，即词汇语义 - WordNet [40], 地理 - GeoNames [1], 生物医学 - 统一医学语言系统 (UMLS) [7] 从美国国家癌症研究所 (NCI) [43], MEDCIN [39], 和系统化医学术语美国 (SNOMEDCT US) [56] 子本体，以及网络中的内容表示 - schema.org [47]。任务A、B和C仅应用于UMLS。换句话说，本体有一个支持知识库，其中的术语可以在测试提示中用于术语类型（任务A）、分类层次关系提示（任务B）和非分类层次关系提示（任务C）。GeoNames来源附带了一个用于类型和分类关系的术语知识库，因此，在本工作的LLM中，它被用于任务A和任务B的本体测试。WordNet来源只能用于任务A，因为它附带了一个用于句法类型的词汇术语实例集合。由于本工作中定义的本体学习任务在任务B和任务C中不适用，因为WordNet中的语义关系是词汇语义的，换句话说，是直接在术语之间，而不是它们的类型之间。最后，由于schema.org来源仅提供作为标准化下载的定型分类，它仅在本文工作的本体学习任务B中被利用。在这种情况下，我们避免了为schema.org分类抓取网络实例。对于本工作中考虑的所有其他与任务A相关的本体知识源，术语实例直接从来源获取。这便于轻松复制我们的任务A数据集。下一节将详细介绍每个任务的本体知识源及相关数据集统计信息。

任务A数据集。 表1展示了任务A数据集的统计信息，我们使用了来自WordNet、GeoNames和UMLS的术语。对于WordNet，我们使用了WN18RRdata dump [14]，它源自原始WordNet，但作为基准数据集发布，包含预创建的训练和测试集。总体而言，它包含40,943个术语，术语之间存在18种不同的关系类型，术语分为四种类型（名词、动词、副词、形容词）。我们将原始的验证集和测试集合并为一个测试数据集。GeoNames包含680类地理位置，这些位置被分为9个更高级别的类别，例如H类（河流、湖泊和海洋），以及R类（道路和铁路）。UMLS包含来自各种来源的近三百万个概念，这些概念通过语义关系相互链接。UMLS的独特之处在于它是一个更大的语义本体网络，包含其他生物医学问题域受限的子本体。

Again, the prefix prompt template reuses the cloze prompt template as the other tasks, with instructions similar to task B. It is $f_{p-prompt}^C(h, r, t) = [instruction] + f_{c-prompt}^C(h, r, t)$

4 LLMs4OL - Three Ontology Learning Tasks Evaluations

4.1 Evaluation Datasets - Ontological Knowledge Sources

To comprehensively assess LLMs for the three OL tasks presented in the previous section, we cover a variety of ontological knowledge domain sources. Generally, across the tasks, four knowledge domains are represented, i.e. lexicosemantic – WordNet [40], geographical – GeoNames [1], biomedicine – Unified Medical Language System (UMLS) [7] teased out as the National Cancer Institute (NCI) [43], MEDCIN [39], and Systematized Nomenclature of Medicine – Clinical Terms United States (SNOMEDCT_US) [56] subontologies, and content representations in the web – schema.org [47]. Tasks A, B, and C applied only to UMLS. In other words, the ontology has a supporting knowledge base with terms that can be leveraged in the test prompts for term typing as Task A, taxonomic hierarchical relational prompts as Task B, and non-taxonomic heterarchical relational prompts as Task C. The GeoNames source came with a knowledge base of terms instantiated for types and taxonomic relations, therefore, was leveraged in the Task A and B as OL tests with LLMs of this work. The WordNet source could be leveraged only in Task A since it came with an instantiated collection of lexical terms for syntactic types. It was not applicable in the Tasks B and C for OL defined in this work since the semantic relations in WordNet are lexicosemantic, in other words, between terms directly and not their types. Finally, since the schema.org source offered only typed taxonomies as standardized downloads, it was leveraged only in the OL Task B of this work. In this case, we refrained from scraping the web for instantiations of the schema.org taxonomy. For all other ontological knowledge sources considered in this work that were relevant to Task A, the term instantiations were obtained directly from the source. This facilitates replicating our Task A dataset easily. Detailed information on the ontological knowledge sources per task with relevant dataset statistics are presented next.

Task A Datasets. Table 1 shows statistical insights for the Task A dataset where we used terms from WordNet, GeoNames, and UMLS. For WordNet we used the WN18RR data dump [14] that is derived from the original WordNet but released as a benchmark dataset with precreated train and test splits. Overall, it consists of 40,943 terms with 18 different relation types between the terms and four term types (noun, verb, adverb, adjective). We combined the original validation and test sets as a single test dataset. GeoNames comprises 680 categories of geographical locations, which are classified into 9 higher-level categories, e.g. H for stream, lake, and sea, and R for road and railroad. UMLS contains almost three million concepts from various sources which are linked together by semantic relationships. UMLS is unique in that it is a greater semantic ontological network that subsumes other biomedical problem-domain restricted subontolo-

表1. 任务A术语类型数据集在三个核心本体知识源（即WordNet、GeoNames和UMLS）上的计数，其中任务A中的UMLS仅由NCI、MEDCIN和SNOMEDCT US亚本体源表示。还提供了定义任务A本体学习的每个源的唯一术语类型。

参数	WordNet	GeoNames	NCI	MEDCIN	SNOMEDCT	美国 -
<i>Train Set Size</i>	40,559	8,078,865	96,177	277,028		278,374
<i>Test Set Size</i>	9,470	702,510	24,045	69,258		69,594
<i>Types</i>	4	680	125	87		125

我们基于三个选定的子本体来源的语义空间，将术语类型任务 grounding 到 NCI、MEDCIN 和 SNOMEDCT US 上。训练数据集用于大语言模型的微调。在11个模型中，我们根据其零样本性能选择了最有前景的一个。测试数据集用于在零样本和微调设置下的评估。

任务B数据集。 从GeoNames、UMLS和schema.org中，我们获得了689、127和797种术语类型，形成了类型分类体系。我们的测试数据集构建为类型对，其中一半表示分类层次结构，另一半不属于任何分类。这是基于以下公式。

$$\forall(a \in T_n, b \in T_{n+1}) \mapsto (aRb \wedge b \neg Ra)$$

$$\forall(a \in T_n, b \in T_{n+1}, c \in T_{n+2}); (aRb \wedge bRc) \mapsto aRc$$

$$\forall(a \in T_n, b \in T_{n+1}, c \in T_{n+2}); (c \neg Rb \wedge b \neg Ra) \mapsto c \neg Ra$$

其中 a 、 b 和 c 是分类层次结构中不同级别的类型。 T 是分类体系中特定级别的一组类型，其中 $n + 2 > n + 1 > n$ 和 n 是根。符号 R 表示“ a 是类型 b 的父类”，作为真实分类关系。相反， $\neg R$ 表示“ b 是类型 a 的父类”，作为虚假分类关系。此外，传递分类关系 $(aRb \wedge bRc) \mapsto aRc$ 也被提取为真实关系，而它们的逆关系，即 $(c \neg Rb \wedge b \neg Ra) \mapsto c \neg Ra$ ，则为虚假关系。

任务C数据集。 如前所述，任务C评估，即非分类关系发现，被限定在我们考虑的仅有的本体知识源中，即UMLS。它报告了在其127个术语类型中存在53个非分类关系。测试数据集包含每个关系的所有类型对，其中对于任何给定的关系，某些对是真实的，而其余的是错误候选。任务B和任务C数据集的统计数据见表2。

4.2 评估模型 - 大语言模型 (LLMs)

如前文所述，在本工作中，我们全面评估了社区中报告的八种主要领域无关的大语言模型，这些模型被视为不同任务的当前最佳技术。它们分别是：BERT [15] 作为仅编码器架构，BLOOM [55]、LLaMA [58]、GPT-3 [9]、GPT-3.5 [45]，以及GPT-4 [46] 作为仅解码器模型，最后是BART [32] 和Flan-T5 [10] 作为编码器-解码器模型。

Table 1. Task A term typing dataset counts across three core ontological knowledge sources, i.e. WordNet, GeoNames, and UMLS, where for Task A UMLS is represented only by the NCI, MEDCIN, and SNOMEDCT_US subontological sources. The unique term types per source that defined Task A Ontology Learning is also provided.

Parameter	WordNet	GeoNames	NCI	MEDCIN	SNOMEDCT_US
<i>Train Set Size</i>	40,559	8,078,865	96,177	277,028	278,374
<i>Test Set Size</i>	9,470	702,510	24,045	69,258	69,594
<i>Types</i>	4	680	125	87	125

gies. We grounded the term typing task to the semantic spaces of three select subontological sources, i.e. NCI, MEDCIN, and SNOMEDCT_US.

The train datasets were reserved for LLM fine-tuning. Among the 11 models, we selected the most promising one based on its zero-shot performance. The test datasets were used for evaluations in both zero-shot and fine-tuned settings. **Task B Datasets.** From GeoNames, UMLS, and schema.org we obtained 689, 127, and 797 term types forming type taxonomies. Our test dataset was constructed as type pairs, where half represented the taxonomic hierarchy while the other half were not in a taxonomy. This is based on the following formulations.

$$\forall(a \in T_n, b \in T_{n+1}) \mapsto (aRb \wedge b \neg Ra)$$

$$\forall(a \in T_n, b \in T_{n+1}, c \in T_{n+2}); (aRb \wedge bRc) \mapsto aRc$$

$$\forall(a \in T_n, b \in T_{n+1}, c \in T_{n+2}); (c \neg Rb \wedge b \neg Ra) \mapsto c \neg Ra$$

Where a , b , and c are types at different levels in the hierarchy. T is a collection of types at a particular level in the taxonomy, where $n + 2 > n + 1 > n$ and n is the root. The symbol R represents “ a is a super class of type b ” as a true taxonomic relation. Conversely, the $\neg R$ represents “ b is a super class of type a ” as a false taxonomic relation. Furthermore, transitive taxonomic relations, $(aRb \wedge bRc) \mapsto aRc$, were also extracted as true relations, while their converse, i.e. $(c \neg Rb \wedge b \neg Ra) \mapsto c \neg Ra$ were false relations.

Task C Datasets. As alluded to earlier, Task C evaluations, i.e. non-taxonomic relations discovery, were relegated to the only available ontological knowledge source among those we considered i.e. UMLS. It reports 53 non-taxonomic relations across its 127 term types. The testing dataset comprised all pairs of types for each relation, where for any given relation some pairs are true while the rest are false candidates. Task B and Task C datasets’ statistics are in Table 2.

4.2 Evaluation Models - Large Language Models (LLMs)

As already introduced earlier, in this work, we comprehensively evaluate eight main types of domain-independent LLMs reported as state-of-the-art for different tasks in the community. They are: BERT [15] as an encoder-only architecture, BLOOM [55], LLaMA [58], GPT-3 [9], GPT-3.5 [45], and GPT-4 [46] as decoder-only models, and finally BART [32] and Flan-T5 [10] as encoder-decoder models.

表2. 数据集统计信息，按任务B类型分类层次结构发现和任务C类型非分类异构结构发现，分别按相关本体知识源统计每个任务报告的参数数量。

任务 参数	GeoNames	UMLS schema.org	
类型	689	127	797
<i>Task B</i> 级别	2	3	6
正向/负向样本	680/680	254/254	2,670/2,670
<i>Train/Test split</i>	272/1,088	101/407	1,086/4,727
非分类关系	-	53	-
<i>Task C</i> 正向/负向样本	-	5,641/1,896	-
<i>Train/Test Split</i>	-	1,507/6,030	-

请注意这些大语言模型以不同的参数规模发布。因此，根据括号中写入的参数大小进行分类，我们总共评估了七种大语言模型：1. BERT-Large (340M), 2. BART-Large (400M), 3. Flan-T5-Large (780M), 4. Flan-T5-XL (3B), 5. BLOOM-1b7 (1.7B), 6. BLOOM-3b (3B), 7. GPT-3 (175B), 8. GPT-3.5 (174B), 9. LLaMA (7B), 以及GPT-4 (>1T)。此外，我们还测试了一种第十一种生物医学领域特定模型PubMedBERT [18]。

在这项工作中，由于我们首次提出了LLMs4OL范式，在某种意义上将本体学习（OL）假定为大语言模型（LLMs）的一种涌现能力，因此对我们来说测试不同的大语言模型在新任务上的表现非常重要。评估不同的大语言模型支持：1) 性能比较——这使我们能够识别哪些模型对本体学习是有效的，2) 模型改进——面向本体学习，人们可以识别模型需要改进的领域，3) 研究进展——通过测试和比较不同的模型，对本体学习感兴趣的研究人员有可能识别新的研究领域，并开发新的技术来改进大语言模型。

4.3 评估

指标。 任务A的评估结果报告为均值平均精度@k（MAP@k），其中 $k = 1$ ，因为该指标被指出最适合该任务。具体来说，在我们的案例中，对于术语类型，MAP@1衡量了由大语言模型（LLM）为使用评估集中术语初始化的提示返回的top-1排名术语类型的平均精度。评估任务B和C的结果则基于精确率和召回率的标准F1分数报告。

结果 - 三个本体学习任务零样本评估。 每个任务的总体评估结果报告在表3中。表格的三个主要行，分别用字母A、B和C标记，对应术语类型、类型分类发现和非分类关系层次发现结果。针对任务A的五个子行显示了WordNet、GeoNames和三个UMLS子本体（即NCI、SNOMEDCT US和MEDCIN）的术语类型结果。针对任务B的三个子行分别显示了GeoNames、UMLS和schema.org的类型分类发现结果。任务C评估结果仅提供UMLS。我们首先考察零样本设置下的结果，即针对三个RQ对大语言模型进行开箱评估的结果。

Table 2. Dataset statistics as counts per reported parameter for Task B type taxonomic hierarchy discovery and Task C type non-taxonomic heterarchy discovery across the pertinent ontological knowledge sources respectively per task.

Task	Parameter	GeoNames	UMLS	schema.org
	Types	689	127	797
<i>Task B</i>	Levels	2	3	6
	Positive/Negative Samples	680/680	254/254	2,670/2,670
	<i>Train/Test split</i>	272/1,088	101/407	1,086/4,727
	Non-Taxonomic Relations	-	53	-
<i>Task C</i>	Positive/Negative Samples	-	5,641/1,896	-
	<i>Train/Test Split</i>	-	1,507/6,030	-

Note these LLMs are released at varying parameter sizes. Thus qualified by the size in terms of parameters written in parenthesis, in all, we evaluate seven LLMs: 1. BERT-Large (340M), 2. BART-Large (400M), 3. Flan-T5-Large (780M), 4. Flan-T5-XL (3B), 5. BLOOM-1b7 (1.7B), 6. BLOOM-3b (3B), 7. GPT-3 (175B), 8. GPT-3.5 (174B), 9. LLaMA (7B), and GPT-4 (>1T). Additionally, we also test an eleventh biomedical domain-specific model PubMedBERT [18].

In this work, since we propose the LLMs4OL paradigm for the first time, in a sense postulating OL as an emergent ability of LLMs, it is important for us to test different LLMs on the new task. Evaluating different LLMs supports: 1) Performance comparison - this allows us to identify which models are effective for OL, 2) Model improvement - toward OL one can identify areas where the models need improvement, and 3) Research advancement - with our results from testing and comparing different models, researchers interested in OL could potentially identify new areas of research and develop new techniques for improving LLMs.

4.3 Evaluations

Metrics. Evaluations for Task A are reported as the mean average precision at k (MAP@K), where $k = 1$, since this metric was noted as being best suited to the task. Specifically, in our case, for term typing, MAP@1 measures the average precision of the top-1 ranked term types returned by an LLM for prompts initialized with terms from the evaluation set. And evaluations for Tasks B and C are reported in terms of the standard F1-score based on precision and recall.

Results - Three Ontology Learning Tasks Zero-shot Evaluations. The per task overall evaluations are reported in Table 3. The three main rows of the table marked by alphabets A, B, and C correspond to term typing, type taxonomy discovery, and type non-taxonomic relational hetrarchy discovery results, respectively. The five subrows against Task A shows term typing results for WordNet, GeoNames, and the three UMLS subontologies, viz. NCI, SNOMEDCT_US, and MEDCIN. The three subrows against Task B shows type taxonomy discovery results for GeoNames, UMLS, and schema.org, respectively. Task C evaluation

表3. 零样本测试在11个大语言模型和微调的Flan-T5-Large及Flan-T5- XL大语言模型上的结果，用于本体学习任务A，即MAP@1中的术语类型，作为任务B（即类型分类发现）的F1分数，以及任务C（即类型非分类关系抽取）的结果。结果以百分比表示。

Task	Dataset	零样本测试											微调	
		BERT-Large	PubMedBERT	BART-Large	Flan-T5-Large	Flan-T5-XL	BLOOM-1b7	BLOOM-3b	GPT-3	GPT-3.5	LLaMA-7B	GPT-4	Flan-T5-Large*	Flan-T5-XL*
A	WordNet	27.9	- 2.2	31.3	52.2	79.2	79.1	37.9	91.7	81.4	90.1	76.9	86.3	
	GeoNames	38.3	- 23.2	13.2	33.8	28.5	28.8	22.4	35.0	29.5	43.3	16.9		18.4
	NCI	11.1	5.9	9.9	9.0	9.8	12.4	15.6	12.7	14.7	7.7	16.1	31.9	32.8
	SNOMEDCT_US	21.1	28.5	19.8	24.3	31.6	37.0	37.7	24.4	25.0	13.8	27.8	33.4	43.4
	MEDCIN	8.7	15.6	12.7	13.0	18.5	28.8	29.8	25.7	23.9	4.9	23.7	38.4	51.8
B	GeoNames	54.5	- 55.4	59.6	52.4	36.7	48.3	53.2	67.8	33.5	55.4	62.5	59.1	
	UMLS	48.2	33.7	49.9	55.3	64.3	38.3	37.5	51.6	70.4	32.3	78.1	53.4	79.3
	schema.org	44.1	- 52.9	54.8	42.7	48.6	51.3	51.0	74.4	33.8	74.3	91.7	91.7	
C	UMLS	40.1	42.7	42.4	46.0	49.5	43.1	42.7	38.8	37.5	20.3	41.3	49.1	53.1

结果仅提供UMLS。我们首先考察零样本设置下的结果，即针对三个RQ对大语言模型进行开箱评估的结果。
RQ1: 大语言模型在任务A（即自动化类型发现）上的效果如何？ 我们根据5个子行与行A（即针对任务A评估的各种本体数据集）的结果来考察这个问题。在五个本体来源中，GPT-3.5在4类型的WordNet上取得了最高的术语类型结果，MAP@1达到91.7%。这种高性能部分归因于WordNet简单的类型空间，仅包含4种类型。然而，在WordNet上评估的其他大语言模型中，尤其是GPT-3，30% MAP@1范围内的分数似乎才是常态，最低的由BART-Large达到2.2%。因此，在WordNet上报告高分的大语言模型应被视为更易于句法类型，无论WordNet的类型空间多么简单。考虑所有本体来源，GeoNames具有最细粒度的类型分类，包含680种类型。尽管如此，在该来源上获得的最佳结果是GPT-4的39.4%，BERT-Large紧随其后，达到38.3%。这比在三个生物医学数据集上的类型评估结果更好。即使是特定领域的PubMedBERT也表现不佳。在这方面，具有大规模参数的领域无关模型（如BLOOM(3B)）更易于处理这项复杂任务。由于生物医学涉及更深的领域特定语义，我们假设不仅可以通过领域特定微调，还可以通过策略性任务特定推理来获得更好的性能。

结果概述是：91.7% WordNet 由 GPT-3.5 > 39.4% GeoNames 由 GPT-4 > 37.7% SNOMEDCT 美国 由 BLOOM-3b -> 29.8% MEDCIN 由 BLOOM-3b > 16.1% NCI 由GPT-4。

Table 3. Zero-shot results across 11 LLMs and finetuned Flan-T5-Large and Flan-T5-XL LLMs results reported for ontology learning Task A i.e. term typing in MAP@1, and as F1-score for Task B i.e. type taxonomy discovery, and Task C i.e. type non-taxonomic relation extraction. The results are in percentages.

Task	Dataset	Zero-Shot Testing											Finetuned	
		BERT-Large	PubMedBERT	BART-Large	Flan-T5-Large	Flan-T5-XL	BLOOM-1b7	BLOOM-3b	GPT-3	GPT-3.5	LLaMA-7B	GPT-4	Flan-T5-Large*	Flan-T5-XL*
A	WordNet	27.9	- 2.2	31.3	52.2	79.2	79.1	37.9	91.7	81.4	90.1	76.9	86.3	
	GeoNames	38.3	- 23.2	13.2	33.8	28.5	28.8	22.4	35.0	29.5	43.3	16.9		18.4
	NCI	11.1	5.9	9.9	9.0	9.8	12.4	15.6	12.7	14.7	7.7	16.1	31.9	32.8
	SNOMEDCT_US	21.1	28.5	19.8	24.3	31.6	37.0	37.7	24.4	25.0	13.8	27.8	33.4	43.4
	MEDCIN	8.7	15.6	12.7	13.0	18.5	28.8	29.8	25.7	23.9	4.9	23.7	38.4	51.8
B	GeoNames	54.5	- 55.4	59.6	52.4	36.7	48.3	53.2	67.8	33.5	55.4	62.5	59.1	
	UMLS	48.2	33.7	49.9	55.3	64.3	38.3	37.5	51.6	70.4	32.3	78.1	53.4	79.3
	schema.org	44.1	- 52.9	54.8	42.7	48.6	51.3	51.0	74.4	33.8	74.3	91.7	91.7	
C	UMLS	40.1	42.7	42.4	46.0	49.5	43.1	42.7	38.8	37.5	20.3	41.3	49.1	53.1

results are provided only for UMLS. We first examine the results in the zero-shot setting, i.e. for LLMs evaluated out-of-the-box, w.r.t. three RQs.
RQ1: How effective are LLMs for Task A, i.e. automated type discovery? We examine this question given the results in 5 subrows against the row A, i.e. corresponding to the various ontological datasets evaluated for Task A. Of the five ontological sources, the highest term typing results were achieved on the 4-typed WordNet at 91.7% MAP@1 by GPT-3.5. This high performance can be attributed in part to the simple type space of WordNet with only 4 types. However, looking across the other LLMs evaluated on WordNet, in particular even GPT-3, scores in the range of 30% MAP@1 seem to be the norm with a low of 2.2% by BART-Large. Thus LLMs that report high scores on WordNet should be seen as more amenable to syntactic typing regardless of the WordNet simple type space. Considering all the ontological sources, Geonames presents the most fine-grained types taxonomy of 680 types. Despite this, the best result obtained on this source is 39.4% from GPT-4 with BERT-Large second at a close 38.3%. This is better than the typing evaluations on the three biomedical datasets. Even the domain-specific PubMedBERT underperforms. In this regard, domain-independent models with large-scale parameters such a BLOOM (3B) are more amenable to this complex task. Since biomedicine entails deeper domain-specific semantics, we hypothesize better performance not just from domain-specific finetuning but also strategically for task-specific reasoning. The results overview is: 91.7% WordNet by GPT-3.5 > 39.4% GeoNames by GPT-4 > 37.7% SNOMEDCT_US by BLOOM-3b > 29.8% MEDCIN by BLOOM-3b > 16.1% NCI by GPT-4.

RQ2: LLMs在识别类型分类法（即类型之间的“is-a”层次结构）方面的效果如何？ 我们根据任务B所评估的三个本体论来源（即三个子行相对于主行B的结果）来检验这个问题。GPT-4在UMLS上取得了最高结果，为78.1%。在开源模型中，Flan-T5-XL取得了最佳结果，为64.3%。因此，对于术语分类法发现，LLMs在生物医学领域的零样本设置中平均表现最为有效。

结果概述是：GPT-4 > 74达到78.1%的UMLS，GPT-3达到4%的schema.org，GPT-3.5达到8%的GeoNames。请注意这三个GPT模型并未开源，因此我们使用付费订阅进行了测试。对于开源模型，结果概述是：Flan-T5-XL达到64.3%的UMLS，Flan-T5-XL达到6%的GeoNames，Flan-T5-Large达到8%的schema.org。

RQ3: 大语言模型发现类型之间非分类关系的效果如何？ 我们根据任务C（即UMLS）的表3行中的结果来考察这个问题。最佳结果是Flan-T5-XL达到49.5%。我们认为这是一个相当好的结果，因为涉及了7,537对类型对，这些类型对要么确实存在非分类关系，要么是错误的配对。

最后，在LLMs4OL范式下考虑的所有三个任务中，术语类型被证明是最难的，其总体结果在大多数测试的其本体论来源中最低，特别是生物医学领域。此外，在我们的分析中，GPT、Flan-T5和BLOOM变体分别随着参数的增加而提高了分数。这一点对闭源的GPT模型也适用，即GPT-3 (175B) 和GPT-3.5 (175B) 到GPT-4 (>1T)以及开源模型，即Flan-T5-Large (780M) 到Flan-T5-XL (3B)和BLOOM从1.7B到3B。因此，似乎很明显，随着大语言模型参数数量的增加，我们可以预期本体学习会有所改进。本体学习。

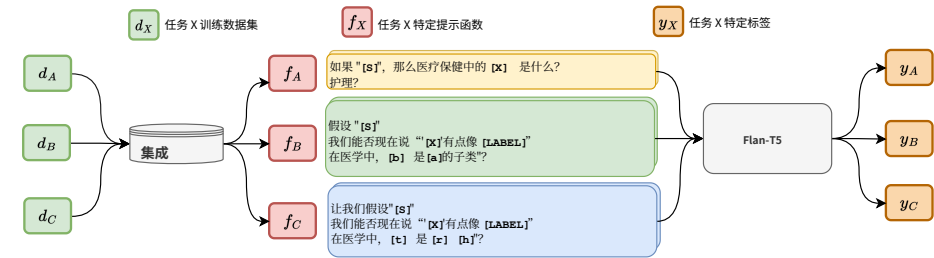


图2。大语言模型在本体学习任务上微调工作流程的示意图。

结果 - 三个本体学习任务微调大语言模型评估。

我们的零样本测试结果表明，虽然大语言模型（LLMs）在本体学习（OL）方面看似有前景，但它们需要针对特定任务的微调才能成为一项实用的解决方案。为此，

RQ2: How effective are LLMs to recognize a type taxonomy i.e. the “is-a” hierarchy between types? We examine this question given the results in the 3 subrows against the main row B, i.e. corresponding to the three ontological sources evaluated for Task B. The highest result was achieved for UMLS by GPT-4 at 78.1%. Of the open-source models, Flan-T5-XL achieved the best result at 64.3%. Thus for the term taxonomy discovery LLMs on average have proven most effective in the zero-shot setting on the biomedical domain.

The results overview is: 78.1% UMLS by GPT-4 > 74.4% schema.org by GPT-3.5 > 67.8% GeoNames by GPT-3.5. Note the three GPT models were not open-sourced and thus we tested them with a paid subscription. For the open-source models, the results overview is: 64.3% UMLS by Flan-T5-XL > 59.6% GeoNames by Flan-T5-XL > 54.8% schema.org by Flan-T5-Large.

RQ3: How effective are LLMs to discover non-taxonomic relations between types? We examine this question given the results in Table 3 row for Task C, i.e. for UMLS. The best result achieved is 49.5% by Flan-T5-XL. We consider this a fairly good result over a sizeable set of 7,537 type pairs that are in true non-taxonomic relations or are false pairs.

Finally, over all the three tasks considered under the LLMs4OL paradigm, term typing proved the hardest obtaining the lowest overall results for most of its ontological sources tested including the biomedical domain in particular. Additionally in our analysis, GPT, Flan-T5, and BLOOM variants showed improved scores with increase in parameters, respectively. This held true for the closed-sourced GPT models, i.e. GPT-3 (175B) and GPT-3.5 (175B) to GPT-4 (>1T) and the open-sourced models, i.e. Flan-T5-Large (780M) to Flan-T5-XL (3B) and BLOOM from 1.7B to 3B. Thus it seems apparent that with an increased number of LLM parameters, we can expect an improvement in ontology learning.

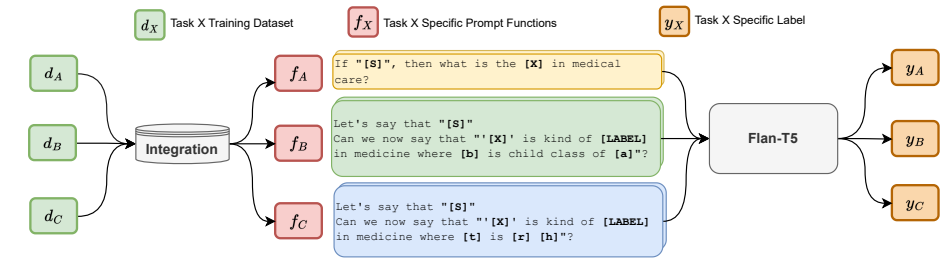


Fig. 2. An illustration of the LLM finetuning workflow on tasks for ontology learning.

Results - Three Ontology Learning Tasks Finetuned LLM Evaluations.

Our zero-shot test results indicate that while LLMs seem promising for OL they would need task-specific finetuning to be a practically viable solution. To this

我们采用了FLAN集合中提出的“指令微调”方法，这是目前唯一已知能够系统性地分解并有效微调大语言模型（LLMs） [35]的方法。为了进行微调，我们选择了Flan-T5大语言模型，原因如下：1) 它是开源的：我们旨在促进未来对未受付费墙遮蔽的模型的研究方向，以帮助实现大语言模型（LLMs）研究的民主化，2) 它在所有任务中始终表现出良好的性能。微调指令是从每个知识源保留的训练集中选取的少量八个样本实例化而来，并输入到图2中所示的微调工作流程中。微调后的Flan模型的结果（参见表格3的最后两列）在几乎所有任务中都显著提升。对于任务A，我们观察到两种Flan-T5变体的微调模型从零样本到微调后的平均改进率为25%。值得注意的是，SNOMEDCT US的改进最少，为9%，而WordNet的改进最多，为45%。对于任务B，我们标记的平均改进率为18%，对于任务C为3%。根据<样式 id='2'>??</样式>中的结果图所示，平均而言，微调后的模型，即使参数更少，在三个本体学习（OL）任务中也优于参数多1000倍或更多的模型。这些见解似乎对于加速使用大语言模型（LLMs）的本体学习（OL）实用工具的开发研究进展至关重要，我们计划在未来的工作中利用这些见解。

5 结论与未来方向

各种举措基准测试大语言模型的性能，揭示了新任务的能力[57,62]。这些基准测试加深了计算机科学界对大语言模型的理解。我们通过引入的概念框架LLMs4OL，探索了大语言模型在本体学习方面的潜力 [17,38] 。在三个本体学习任务上对11个大语言模型进行的广泛实验，证明了该范式的概念验证。我们的代码库便于复制和扩展用于测试新大语言模型的方法。我们的实证结果很有前景，为未来本体学习领域的工作铺平了道路。

在基于大语言模型的本体学习领域，未来的研究方向可以聚焦于几个关键方面。首先，需要针对本体学习任务增强大语言模型，探索新型架构并进行微调以更好地捕捉本体结构。其次，将评估扩展到当前工作中所考察的多样化知识领域之外，将有助于更全面地理解大语言模型的泛化能力。第三，结合大语言模型与传统本体学习技术（如词汇句法模式挖掘和聚类）的混合方法，可能产生更准确、更全面的本体。第四，进一步研究可以深入到特定语义关系的提取，如部分整体关系或因果关系，以增强学习本体的表达能力。标准化评估指标、创建基准数据集、探索动态本体演化以及领域特定学习都是重要的研究方向。此外，将人机交互方法与专家参与相结合，将提高本体的相关性和准确性。探索这些研究方向将推进基于大语言模型的本体学习，增强跨领域的知识获取和表示能力。

end, we adopt the method of “instruction tuning” proposed as the FLAN collection which is the only known systematically deconstructed, effective way to finetune LLMs [35]. For finetuning, we choose the Flan-T5 LMM for two reasons: 1) it is open-source: we intend to foster future research directions for models unhidden behind paywalls to aid in democratizing LLM research, and 2) it showed consistently good performance across all tasks. The finetuning instructions were instantiated from a small selection of eight samples of each knowledge source’ reserved training set and fed in a finetuning workflow shown in Figure 2. The finetuned Flan models’ results (see last two columns in Table 3) are significantly boosted across almost all tasks. For task A, we observed an average improvement of 25% from zero-shot to the finetuned model for both Flan-T5 variants. Notably, SNOMEDCT.US showed least improvement of 9%, while the WordNet showed the most improvement of 45%. For task B we marked an average improvement of 18%, and for task C 3%. Given an illustration of the results in ?? shows that on average finetuned models, even with fewer parameters outperforms models with 1000x or more parameters across the three OL tasks. These insights appear crucial to expedite developmental research progress for practical tools for OL using LLMs which we plan to leverage in our future work.

5 Conclusions and Future Directions

Various initiatives benchmark LLM performance, revealing new task abilities [57,62]. These benchmarks advance computer science’s understanding of LLMs. We explore LLMs’ potential for Ontology Learning [17,38] through our introduced conceptual framework, LLMs4OL. Extensive experiments on 11 LLMs across three OL tasks demonstrate the paradigm’s proof of concept. Our code-base facilitates replication and extension of methods for testing new LLMs. Our empirical results are promising to pave future work for OL.

Future research directions in the field of OL with LLMs can focus on several key areas. First, there is a need to enhance LLMs specifically for ontology learning tasks, exploring novel architectures and fine-tuning to capture ontological structures better. Second, expanding the evaluation to cover diverse knowledge domains beyond the ones examined in the current work would provide a broader understanding of LLMs’ generalizability. Third, hybrid approaches that combine LLMs with traditional ontology learning techniques, such as lexico-syntactic pattern mining and clustering, could lead to more accurate and comprehensive ontologies. Fourth, further research can delve into the extraction of specific semantic relations, like part-whole relationships or causality, to enhance the expressiveness of learned ontologies. Standardizing evaluation metrics, creating benchmark datasets, exploring dynamic ontology evolution, and domain-specific learning are important directions. Additionally, integrating human-in-the-loop approaches with expert involvement would enhance ontology relevance and accuracy. Exploring these research directions will advance LLM-based ontology learning, enhancing knowledge acquisition and representation across domains.

补充材料声明： 我们的LLM模板、详细结果和代码库已作为补充材料公开发布在Github上 <https://github.com/HamedBabaei/LLMs4OL>。

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Supplemental Material Statement: Our LLM templates, detailed results, and codebase are publicly released as supplemental material on Github <https://github.com/HamedBabaei/LLMs4OL>.

Author Contributions

Hamed Babaei Giglou: Conceptualization, Methodology, Software, Validation, Investigation, Resources, Data Curation, Writing - Original Draft, Visualization. Jennifer D’Souza: Conceptualization, Methodology, Investigation, Resources, Writing - Original Draft, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. Sören Auer: Conceptualization, Methodology, Investigation, Resources, Review & Editing, Supervision, Project administration, Funding acquisition.

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附录A

A.1 任务A

WordNet数据集的提示模板。 用于WordNet在任务A的零样本测试中的模板在表4中给出。

GeoNames数据集的提示模板。 用于GeoNames在任务A的零样本测试中的模板在表5中给出。作为一句 S , 我们使用了 $[L]$ *is a place in* $[COUNTRY]$. 模板。

UMLS数据集的提示模板。 用于UMLS来源（NCI、MEDCIN和SNOMEDCT US）在任务A的零样本测试中的模板在表6中给出。

A.2 任务B

用于 GeoNames、UMLS 和 Schema.Org 在任务B的零样本测试中的模板, 为大语言模型呈现于表7。

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A Apendix

A.1 Task A

Prompt Templates for WordNet Dataset. Templates for WordNet in zero-shot testing for Task A are presented in Table 4.

Prompt Templates for GeoNames Dataset. Templates for GeoNames in zero-shot testing for Task A are presented in Table 5. As a sentence S , we used $[L]$ *is a place in* $[COUNTRY]$. template.

Prompt Templates for UMLS Dataset. Templates for UMLS sources (NCI, MEDCIN, and SNOMEDCT-US) in zero-shot testing for Task A are presented in Table 6.

A.2 Task B

Templates for GeoNames, UMLS, and Schema.Org in zero-shot testing for Task B is for LLMs are presented in Table 7.

表4. WordNet零样本测试提示模板用于任务A。 L 表示词汇条目， S 表示包含 L 的句子。在BERT/BART大语言模型中，对于BART， $[MASK]$ 正被替换为 $< mask >$ 。”

LLMs	Prompt Templates
BERT/BART	[S]。 [L] POS是一个 $[MASK]$ 。
	[S]。 [L] 词性是一个 $[MASK]$ 。
	[S]。 ’ [L]’ POS是一个 $[MASK]$ 。
	[S]。 ’ [L]’ 词性是一个 $[MASK]$ 。
	[L] 词性是一个 $[MASK]$ 。
	[L] 词性是一个 $[MASK]$ 。
	’ [L]’ 词性是一个 $[MASK]$ 。
Flan-T5	[S]。 [L] 词性是一个？
	[S]。 [L] 词性是一个？
	[S]。 ’ [L]’ 词性是一个？
	[L] POS 是一个？
	[L] 词性是一个？
	’ [L]’ 词性是一个？
	’ [L]’ 词性是一个？
BLOOM/GPT-3	对以下句子进行句子补全： \n 句子： [S]. [L] 词性是一个
	对以下句子进行句子补全： \n 句子： [S]. [L] 词性是一个
	对以下句子进行句子补全： \n 句子： [S]. ’[L]’ 词性是一个
	对以下句子进行句子补全： \n 句子： [S]。 ’ [L]’ 的词性是
	对以下句子进行句子补全： \n 句子： [L]POS是
	对以下句子进行句子补全： \n 句子： [L]词性是
	对以下句子进行句子补全： \n 句子： ’ [L]’ 词性是
	对以下句子进行句子补全： \n 句子： ’ [L]’ 词性是
	对以下句子进行句子补全： [S]。 [L] 词性是 ____。 \n 答案是
	对以下句子进行句子补全： [S]。 [L] 词性是一个 ____。 \n 答案是
	对以下句子进行句子补全： [S]。 ’ [L]’ 词性是一个 ____。 \n 答案是
	对以下句子进行句子补全： [S]. ’ [L]’ 词性是一个 ____。 \n 答案是
	对以下句子进行句子补全： [L] 词性是一个 ____。 \n 答案是
	对以下句子进行句子补全： [L] 词性是一个 ____。 \n 答案是
LLaMA	对以下句子进行句子补全： ’ [L]’ POS是一个 ____。 \n 答案是
	对以下句子进行句子补全： ’ [L]’ 词性是一个 ____。 \n 答案是

Table 4. The WordNet zero-shot testing prompt templates for task A. L represents lexical entries, S represents sentence containing L . In the BERT/BART LLMs, for BART the $[MASK]$ is being replaced by $< mask >$ ”.

LLMs	Prompt Templates
BERT/BART	[S]. [L] POS is a $[MASK]$.
	[S]. [L] part of speech is a $[MASK]$.
	[S]. ’[L]’ POS is a $[MASK]$.
	[S]. ’[L]’ part of speech is a $[MASK]$.
	[L] POS is a $[MASK]$.
	[L] part of speech is a $[MASK]$.
	’[L]’ POS is a $[MASK]$.
Flan-T5	’[L]’ part of speech is a $[MASK]$.
	[S]. [L] POS is a ?
	[S]. [L] part of speech is a ?
	[S]. ’[L]’ POS is a ?
	[S]. ’[L]’ part of speech is a ?
	[L] POS is a ?
	[L] part of speech is a ?
BLOOM/GPT-3	’[L]’ POS is a ?
	’[L]’ part of speech is a ?
	Perform a sentence completion on the following sentence: \n Sentence: [S]. [L] POS is a
	Perform a sentence completion on the following sentence: \n Sentence: [S]. [L] part of speech is a
	Perform a sentence completion on the following sentence: \n Sentence: [S]. ’[L]’ POS is a
	Perform a sentence completion on the following sentence: \n Sentence: [S]. ’[L]’ part of speech is a
	Perform a sentence completion on the following sentence: \n Sentence: [L] POS is a
	Perform a sentence completion on the following sentence: \n Sentence: [L] part of speech is a
	Perform a sentence completion on the following sentence: \n Sentence: ’[L]’ POS is a
	Perform a sentence completion on the following sentence: \n Sentence: ’[L]’ part of speech is a
	Perform a sentence completion on the following sentence: [S]. [L] POS is a ____。 \n The answer is
	Perform a sentence completion on the following sentence: [S]. [L] part of speech is a ____。 \n The answer is
	Perform a sentence completion on the following sentence: [S]. ’[L]’ POS is a ____。 \n The answer is
	Perform a sentence completion on the following sentence: [S]. ’[L]’ part of speech is a ____。 \n The answer is
LLaMA	Perform a sentence completion on the following sentence: [L] POS is a ____。 \n The answer is
	Perform a sentence completion on the following sentence: [L] part of speech is a ____。 \n The answer is
	Perform a sentence completion on the following sentence: ’[L]’ POS is a ____。 \n The answer is
	Perform a sentence completion on the following sentence: ’[L]’ part of speech is a ____。 \n The answer is

表5. 用于任务A的GeoNames零样本测试提示模板。 L 代表词汇条目， S 代表包含 L 的句子。在BERT/BART大语言模型中，对于BART， $[MASK]$ 正被替换为 $< mask >$ ”。

LLMs	提示模板
<i>BERT/BART</i>	[S]. [L] 是一个 $[MASK]$ 。
	[S]. [L] 地理上是一个 $[MASK]$ [S]. ’ [L]’ 是一个 $[MASK]$. [S]. ’ [L]’ 地理上是一个 $[MASK]$ 。
<i>Flan-T5</i>	[L] 是一个 $[MASK]$ 。
	[L] 地理上是一个 $[MASK]$ 。
<i>BLOOM/GPT-3</i>	’ [L]’ 是一个 $[MASK]$ 。
	[S]. [L] 是一个?
<i>LLaMA</i>	[S]. [L] 在地理上是一个?
	[S]. ’ [L]’ 是一个 ?
<i>BLOOM/GPT-3</i>	[S]. ’ [L]’ 在地理上是一个?
	[L] 是一个?
<i>LLaMA</i>	[L] 在地理上是一个?
	’ [L]’ 是一个?
<i>BLOOM/GPT-3</i>	’ [L]’ 在地理上是一个?
	对以下句子进行句子补全: \n
<i>LLaMA</i>	句子: [S]. [L] 是一个
	对以下句子进行句子补全: \n
<i>BLOOM/GPT-3</i>	句子: [S]. [L] 在地理上是一个
	对以下句子进行句子补全: \n
<i>LLaMA</i>	句子: [S]. ’ [L]’ 是一个
	对以下句子进行句子补全: \n
<i>BLOOM/GPT-3</i>	句子: [S]. ’ [L]’ 从地理位置上是一个
	对以下句子进行句子补全: \n
<i>LLaMA</i>	句子: [L]是一个
	对以下句子进行句子补全: \n
<i>BLOOM/GPT-3</i>	句子: [L]在地理上是
	对以下句子执行句子补全: \n
<i>LLaMA</i>	句子: ’ [L]’ 是
	对以下句子执行句子补全: \n
<i>BLOOM/GPT-3</i>	句子: ’ [L]’ 在地理上是
	对以下句子执行句子补全: [S]. [L]
<i>LLaMA</i>	is a _____. \n 答案是
	对以下句子进行句子补全: [S]. [L]
<i>BLOOM/GPT-3</i>	geographically 是一个 _____. \n 答案是
	对以下句子进行句子补全: [S].
<i>LLaMA</i>	’ [L]’ 是一个____. \n 答案是
	对以下句子进行句子补全: \n [S].
<i>BLOOM/GPT-3</i>	’ [L]’ 从地理位置上讲是 _____. \n 答案是
	对以下句子进行句子补全: \n [L]
<i>LLaMA</i>	is a _____. \n 答案是
	对以下句子进行句子补全: \n [L]
<i>BLOOM/GPT-3</i>	geographically 是一个 _____. \n 答案是
	对以下句子进行句子补全: \n ’ [L]’
<i>LLaMA</i>	is a _____. \n 答案是
	对以下句子进行句子补全: \n ’ [L]’
<i>BLOOM/GPT-3</i>	地理上是一个 _____. \n 答案是

Table 5. The GeoNames zero-shot testing prompt templates for task A. L represents lexical entries, S represents sentence containing L . In the BERT/BART LLMs, for BART the $[MASK]$ is being replaced by $< mask >$ ”.

LLMs	Prompt Templates
<i>BERT/BART</i>	[S]. [L] is a $[MASK]$.
	[S]. [L] geographically is a $[MASK]$. [S]. ’[L]’ is a $[MASK]$. [S]. ’[L]’ geographically is a $[MASK]$. [L] is a $[MASK]$. [L] geographically is a $[MASK]$. ’[L]’ is a $[MASK]$. ’[L]’ geographically is a $[MASK]$.
<i>Flan-T5</i>	[S]. [L] is a ?
	[S]. [L] geographically is a ? [S]. ’[L]’ is a ? [S]. ’[L]’ geographically is a ? [L] is a ? [L] geographically is a ? ’[L]’ is a ? ’[L]’ geographically is a ?
<i>BLOOM/GPT-3</i>	Perform a sentence completion on the following sentence: \n
	Sentence: [S]. [L] is a
<i>LLaMA</i>	Perform a sentence completion on the following sentence: \n
	Sentence: [S]. [L] geographically is a
<i>BLOOM/GPT-3</i>	Perform a sentence completion on the following sentence: \n
	Sentence: [S]. ’[L]’ is a
<i>LLaMA</i>	Perform a sentence completion on the following sentence: \n
	Sentence: [S]. ’[L]’ geographically is a
<i>BLOOM/GPT-3</i>	Perform a sentence completion on the following sentence: \n
	Sentence: [L] is a
<i>LLaMA</i>	Perform a sentence completion on the following sentence: \n
	Sentence: [L] geographically is a
<i>BLOOM/GPT-3</i>	Perform a sentence completion on the following sentence: \n
	Sentence: ’[L]’ is a
<i>LLaMA</i>	Perform a sentence completion on the following sentence: \n
	Sentence: ’[L]’ geographically is a
<i>BLOOM/GPT-3</i>	Perform a sentence completion on the following sentence: [S]. [L]
	is a _____. \n The answer is
<i>LLaMA</i>	Perform a sentence completion on the following sentence: [S]. [L]
	geographically is a _____. \n The answer is
<i>BLOOM/GPT-3</i>	Perform a sentence completion on the following sentence: [S].
	’[L]’ is a _____. \n The answer is
<i>LLaMA</i>	Perform a sentence completion on the following sentence: \n [S].
	’[L]’ geographically is a _____. \n The answer is
<i>BLOOM/GPT-3</i>	Perform a sentence completion on the following sentence: \n [L]
	is a _____. \n The answer is
<i>LLaMA</i>	Perform a sentence completion on the following sentence: \n [L]
	geographically is a _____. \n The answer is
<i>BLOOM/GPT-3</i>	Perform a sentence completion on the following sentence: \n ’[L]’
	is a _____. \n The answer is
<i>LLaMA</i>	Perform a sentence completion on the following sentence: \n ’[L]’
	geographically is a _____. \n The answer is

表6. UMLS零样本测试提示模板用于任务A. L 表示词汇条目, S 表示包含 L 的句子。在BERT/BART大语言模型中, 对于BART, $[MASK]$ 正被替换为 $< mask > .$ ”

LLMs	提示模板
<i>BERT/BART</i>	[S]. [L] 在医学上是一种 $[MASK]$ 。
	[S]. [L] 在生物医学上是一种 $[MASK]$ 。
	[S]. '[L]' 在医学上是一种 $[MASK]$ 。
	[S]. 在生物医学中, '[L]' 是一个 $[MASK]$ 。
	[L] 在医学中是一个 $[MASK]$ 。
	[L] 在生物医学中是一个 $[MASK]$ 。
	'[L]' 是一个 $[MASK]$ 。
<i>Flan-T5</i>	'[L]' 在生物医学中是一个 $[MASK]$ 。
	[S]. [L] 在医学中是一个?
	[S]. [L] 在生物医学中是一个?
	[S]. '[L]' 在医学中是一个?
	[S]. '[L]' 在生物医学中是一个?
	[L] 在医学中是一个?
	[L] 在生物医学中是一个?
<i>BLOOM/GPT-3</i>	'[L]' 在医学中是一个?
	'[L]' 在生物医学中是一个?
	对以下句子执行句子补全: \n 句子: [S]. [L] 在医学中是一个
	对以下句子执行句子补全: \n 句子: [S]. [L] 在生物医学中是一个
	对以下句子执行句子补全: \n 句子: [S]. '[L]' 在医学中是一种
	对以下句子进行句子补全: \n 句子: [S]. '[L]' 在生物医学中是一种
	对以下句子进行句子补全: \n 句子: [L]在医学中是一种
<i>LLaMA</i>	对以下句子进行句子补全: \n 句子: [L]在生物医学中是一个
	对以下句子进行句子补全: \n 句子: '[L]' 在医学中是一个
	对以下句子进行句子补全: \n 句子: '[L]' 在生物医学中是一个
	对以下句子进行句子补全: Sen- 句子: [S]. [L] 在医学上是一种 _____. \n 答案是
	对以下句子进行句子补全: [S]. [L] 在生物医学上是一种 _____. \n 答案是
	对以下句子进行句子补全: [S]. '[L]' 在医学中是一种 _____. \n 答案是
	对以下句子进行句子补全: [S]. '[L]' 在生物医学中是一种 _____. \n 答案是
<i>LLaMA</i>	对以下句子进行句子补全: [L]in 医学是 _____. \n 答案是
	对以下句子进行句子补全: [L]in 生物医学是 _____. \n 答案是
	对以下句子进行句子补全: '[L]' 在医学中是一个 _____. \n 答案是
	对以下句子进行句子补全: '[L]' 在生物医学中是一个 _____. \n 答案是
	对以下句子进行句子补全: [L] in medicine is a _____. \n The answer is
	对以下句子进行句子补全: [L] in biomedicine is a _____. \n The answer is
	对以下句子进行句子补全: '[L]' in medicine is a _____. \n The answer is

Table 6. The UMLS zero-shot testing prompt templates for task A. L represents lexical entries, S represents sentence containing L . In the BERT/BART LLMs, for BART the $[MASK]$ is being replaced by $< mask > .$

LLMs	Prompt Templates
<i>BERT/BART</i>	[S]. [L] in medicine is a $[MASK]$.
	[S]. [L] in biomedicine is a $[MASK]$.
	[S]. '[L]' in medicine is a $[MASK]$.
	[S]. '[L]' in biomedicine is a $[MASK]$.
	[L] in medicine is a $[MASK]$.
	[L] in biomedicine is a $[MASK]$.
	'[L]' is a $[MASK]$.
<i>Flan-T5</i>	'[L]' in biomedicine is a $[MASK]$.
	[S]. [L] in medicine is a ?
	[S]. [L] in biomedicine is a ?
	[S]. '[L]' in medicine is a ?
	[S]. '[L]' in biomedicine is a ?
	[L] in medicine is a ?
	[L] in biomedicine is a ?
<i>BLOOM/GPT-3</i>	'[L]' in medicine is a ?
	'[L]' in biomedicine is a ?
	Perform a sentence completion on the following sentence: \n Sentence: [S]. [L] in medicine is a
	Perform a sentence completion on the following sentence: \n Sentence: [S]. [L] in biomedicine is a
	Perform a sentence completion on the following sentence: \n Sentence: [S]. '[L]' in medicine is a
	Perform a sentence completion on the following sentence: \n Sentence: [S]. '[L]' in biomedicine is a
	Perform a sentence completion on the following sentence: \n Sentence: [L] in medicine is a
<i>LLaMA</i>	Perform a sentence completion on the following sentence: \n Sentence: [L] in biomedicine is a
	Perform a sentence completion on the following sentence: \n Sentence: '[L]' in medicine is a
	Perform a sentence completion on the following sentence: \n Sentence: '[L]' in biomedicine is a
	Perform a sentence completion on the following sentence: Sen- tence: [S]. [L] in medicine is a _____. \n The answer is
	Perform a sentence completion on the following sentence: [S]. [L] in biomedicine is a _____. \n The answer is
	Perform a sentence completion on the following sentence: [S]. '[L]' in medicine is a _____. \n The answer is
	Perform a sentence completion on the following sentence: [S]. '[L]' in biomedicine is a _____. \n The answer is
<i>LLaMA</i>	Perform a sentence completion on the following sentence: [L] in medicine is a _____. \n The answer is
	Perform a sentence completion on the following sentence: [L] in biomedicine is a _____. \n The answer is
	Perform a sentence completion on the following sentence: '[L]' in medicine is a _____. \n The answer is
	Perform a sentence completion on the following sentence: '[L]' in biomedicine is a _____. \n The answer is
	Perform a sentence completion on the following sentence: [L] in medicine is a _____. \n The answer is
	Perform a sentence completion on the following sentence: [L] in biomedicine is a _____. \n The answer is
	Perform a sentence completion on the following sentence: '[L]' in medicine is a _____. \n The answer is

表7。GeoNames、UMLS和Schema.Org的任务B零样本测试提示模板。在类型对 (a, b) 或 (b, a) 中, 其中 a 是超类, b 是子类。

LLMs	提示模板
<i>BERT</i>	[a] 是 [b]的超类。该陈述是一个 $[MASK]$ 。
	[b] 是 [a]的子类。该陈述是一个 $[MASK]$ 。
	[a] 是 [b]的父类。该陈述是一个 $[MASK]$ 。
	[b] 是 [a]的子类。这个陈述是一个 $[MASK]$ 。
	[a] 是 [b]的超类型。这个陈述是一个 $[MASK]$ 。
	[b] 是 [a]的子类型。这个陈述是一个 $[MASK]$ 。
	[a] 是 [b]的祖先类。这个陈述是一个 $[MASK]$ 。
	[b] 是 [a]的后代类。这个陈述是一个 $[MASK]$ 。
<i>BART</i>	[a] 是 [b]的超类。这个陈述是一个 $< mask >$ 。
	[b] 是 [a]的子类。这个陈述是一个 $< mask >$ 。
	[a] 是 [b]的父类。这个陈述是一个 $< mask >$ 。
	[b] 是 [a]的子类。这个陈述是一个 $< mask >$ 。
	[a] 是 [b]的超类型。这个陈述是一个 $< mask >$ 。
	[b] 是 [a]的子类型。这个陈述是一个 $< mask >$ 。
	[a] 是 [b]的祖先类。这个陈述是一个 $< mask >$ 。
	[b] 是 [a]的后代类。这个陈述是一个 $< mask >$ 。
<i>Flan-T5</i>	[a] 是 [b]的超类。这个陈述是一个
	[b] 是 [a]的子类。这个陈述是一个
	[a] 是 [b]的父类。这个陈述是一个
	[b] 是 [a]的子类。这个陈述是一个
	[a] 是 [b]的超类型。这个陈述是一个
	[b] 是 [a]的子类型。这个陈述是一个
	[a] 是 [b]的祖先类。这个陈述是一个
	[b] 是 [a]的后代类。这个陈述是一个
<i>BLOOM/GPT-3</i>	判断以下陈述是否为真: \n
	陈述: [a]是 [b]的超类。 \n这个陈述是一个
	判断以下陈述是否为真: \n
	声明: [b] 是 [a]的子类。 \n 该声明是
	判断以下声明是否为真或假: \n
	声明: [a] 是 [b]的父类。 \n 该声明是
	a
	判断以下声明是否为真或假: \n
	声明: [b]是 [a]的子类。 \n 该声明是
	判断以下声明是否为真或假: \n
	声明: [a]是 [b]的超类型。 \n该声明是
	判断以下声明是否为真或假: \n
	声明: [b] 是 [a]的子类型。 \n该声明是
	判断以下声明是否为真或假: \n
	声明: [a] 是 [b]的祖先类。 \n该声明是
	判断以下声明是否为真或假: \n
	声明: [b] 是 [a]的后代类。 \n 这个声明
	is a

Table 7. The GeoNames, UMLS, and Schema.Org zero-shot testing prompt templates for task B. In the type pairs (a, b) or (b, a) , where a is parent and b is child.

LLMs	Prompt Templates
<i>BERT</i>	[a] is the superclass of [b]. This statement is a $[MASK]$.
	[b] is a subclass of [a]. This statement is a $[MASK]$.
	[a] is the parent class of [b]. This statement is a $[MASK]$.
	[b] is a child class of [a]. This statement is a $[MASK]$.
	[a] is a supertype of [b]. This statement is a $[MASK]$.
	[b] is a subtype of [a]. This statement is a $[MASK]$.
	[a] is an ancestor class of [b]. This statement is a $[MASK]$.
	[b] is a descendant class of [a]. This statement is a $[MASK]$.
<i>BART</i>	[a] is the superclass of [b]. This statement is a $< mask >$.
	[b] is a subclass of [a]. This statement is a $< mask >$.
	[a] is the parent class of [b]. This statement is a $< mask >$.
	[b] is a child class of [a]. This statement is a $< mask >$.
	[a] is a supertype of [b]. This statement is a $< mask >$.
	[b] is a subtype of [a]. This statement is a $< mask >$.
	[a] is an ancestor class of [b]. This statement is a $< mask >$.
	[b] is a descendant class of [a]. This statement is a $< mask >$.
<i>Flan-T5</i>	[a] is the superclass of [b]. This statement is a
	[b] is a subclass of [a]. This statement is a
	[a] is the parent class of [b]. This statement is a
	[b] is a child class of [a]. This statement is a
	[a] is a supertype of [b]. This statement is a
	[b] is a subtype of [a]. This statement is a
	[a] is an ancestor class of [b]. This statement is a
	[b] is a descendant class of [a]. This statement is a
<i>BLOOM/GPT-3</i>	Identify whether the following statement is true or false: \n
	Statement: [a] is the superclass of [b]. \n This statement is a
	Identify whether the following statement is true or false: \n
	Statement: [b] is a subclass of [a]. \n This statement is a
	Identify whether the following statement is true or false: \n
	Statement: [a] is the parent class of [b]. \n This statement is
	a
	Identify whether the following statement is true or false: \n
	Statement: [b] is a child class of [a]. \n This statement is a
	Identify whether the following statement is true or false: \n
	Statement: [a] is a supertype of [b]. \n This statement is a
	Identify whether the following statement is true or false: \n
	Statement: [b] is a subtype of [a]. \nThis statement is a
	Identify whether the following statement is true or false: \n
	Statement: [a] is an ancestor class of [b]. \n This statement is a
	Identify whether the following statement is true or false: \n
	Statement: [b] is a descendant class of [a]. \n This statement
	is a

B Flan-T5 训练设置和超参数

我们在三个任务上微调 Flan-T5 模型，并使用零样本测试在所有三个任务上进行评估。这涉及使用不同的数据源，即 WordNet（任务A）、GeoNames（任务A和B）、UMLS（代表任务A、B和C中医疗来源的NCI数据源）以及 schema.org（任务B）。考虑到任务A作为8-shot实例进行训练，我们将任务B和C的训练样本结合起来，条件是只有那些在任务A 8-shot实例中的样本才被考虑纳入。接下来，使用任务特定的提示模板生成 Flan-T5 输入进行微调。随后，使用设计的提示模板对 Flan-T5 进行微调。

我们对所有数据集和模型使用了相同的训练策略，除了几个超参数：批大小和微调步数。所有模型都使用 AdamW 优化器，学习率为 1e-5 进行训练。对于 Flan-T5- Large 模型，训练期间使用批大小为 8，而对于 Flan-T5- XL 模型，在所有数据集上都使用批大小为 4。WordNet 和 Schema.Org 数据集在两个模型上都微调了 5 个训练周期，同样，UMLS 使用 10 个周期进行微调，而 GeoNames 在 Flan-T5-Large 上微调了 10 个周期，在 Flan-T5-XL 上微调了 6 个周期。

C 详细结果

表8和表9表示针对术语类型，所有模板和大语言模型（LLMs）的提示模板结果。而表10表示针对分类法发现，所有模板和大语言模型（LLMs）的提示模板结果。

B Flan-T5 Training Setups and Hyperparameters

We finetune Flan-T5 LM on three tasks and evaluate them on all three tasks using zero-shot testing. It involved employing different sources, i.e. WordNet (task A), GeoNames (task A and B), UMLS (the NCI source representing medical sources in task A, B, and C), and schema.org (task B). Considering task A as an 8-shot instance for training we combined samples task B and C training with the condition that only samples that are in the task A 8-shot instances are considered for inclusion. Next, using task-specific prompt templates, Flan-T5 inputs are generated for finetuning. Following, using designed prompt templates Flan-T5 is fine-tuned.

We utilized a consistent training strategy for all datasets and models, except for a few hyperparameters: batch size and finetuning steps. All the models were trained using AdamW optimizer with a learning rate of 1e-5. For the Flan-T5- Large model, a batch size of 8 is used during training, while for the Flan-T5- XL model, a batch size of 4 is employed on all datasets. The WordNet and Schema.Org datasets were finetuned for 5 training epochs on both models, similarly, UMLS was finetuned using 10 epochs, while GeoNames was finetuned on Flan-T5-Large for 10 epochs and on Flan-T5-XL for 6 epochs.

C Detailed Results

The Table 8 and Table 9 represent prompt template results across all the templates and LLMs for term typing. While the Table 10 represents prompt template results across all the templates and LLMs for taxonomy discovery.

表8。 零样本测试和微调的详细结果，涵盖七个用于本体学习任务A、MAP@1中的术语类型的大语言模型。结果以百分比表示。 * 表示微调模型结果。WordNet和GeoName数据集的结果。

数据集 大语言模型		提示模板							
		t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8
WordNet	BERT-Large	2.19	9.36	9.18	19.41	4.72	19.34	9.93	27.85
	BART-Large	0.01	0.28	0.22	2.16	0.01	0.03	0.0	0.19
	Flan-T5-Large	0.17	19.70	5.54	31.26	0.0	3.03	5.70	26.80
	BLOOM-1b7	66.83	71.53	79.20	76.84	40.08	61.96	68.39	70.03
	Flan-T5-XL	2.81	40.26	17.83	52.21	0.01	7.75	18.47	18.85
	BLOOM-3b	63.33	75.29	79.08	77.06	37.40	65.32	68.99	71.62
	LLaMA-7B	37.26	74.61	70.16	75.97	24.28	76.61	67.41	81.38
	GPT-3	15.32	26.55	37.86	27.57	8.47	27.13	27.51	24.65
	GPT-3.5	24.27	80.81	89.46	91.72	0.81	60.76	49.38	82.41
	GPT-4	-	-	-	90.11	-	-	-	-
	<i>Flan-T5-Large</i> *	73.32	76.74	54.57	76.90	10.83	61.36	54.29	69.32
	<i>Flan-T5-XL</i> *	84.51	84.77	77.27	86.28	50.23	76.46	72.38	80.51
	BERT-Large	38.34	29.79	30.86	35.32	23.61	25.66	11.32	30.44
	BART-Large	8.47	0.57	2.23	0.98	21.48	20.51	7.83	23.21
GeoNames	Flan-T5-Large	11.55	3.57	13.16	4.68	9.45	6.05	8.17	7.38
	BLOOM-1b7	2.71	2.54	2.89	3.20	28.51	18.38	25.86	19.80
	Flan-T5-XL	33.81	15.71	19.77	20.78	15.36	12.41	18.43	15.82
	BLOOM-3b	3.76	4.70	2.64	3.43	28.84	18.08	25.64	20.71
	LLaMA-7B	29.49	14.16	25.54	15.95	13.91	9.44	17.79	16.79
	GPT-3	22.42	8.72	-	7.50	-	-	-	-
	GPT-3.5	35.00	-	-	-	-	-	-	-
	GPT-4	43.28	-	-	-	-	-	-	-
	<i>Flan-T5-Large</i> *	15.08	15.17	14.93	15.12	15.77	16.28	15.93	16.91
	<i>Flan-T5-XL</i> *	18.35	18.12	18.12	17.91	17.26	17.32	17.45	17.64
	BERT-Large	38.34	29.79	30.86	35.32	23.61	25.66	11.32	30.44
	BART-Large	8.47	0.57	2.23	0.98	21.48	20.51	7.83	23.21
	Flan-T5-Large	11.55	3.57	13.16	4.68	9.45	6.05	8.17	7.38
	BLOOM-1b7	2.71	2.54	2.89	3.20	28.51	18.38	25.86	19.80

Table 8. The detailed results of zero-shot testing and finetuning across seven LLMs reported for ontology learning Task A, term typing in MAP@1. The results are in percentages. The * denotes finetuning model results. Results for WordNet and GeoName datasets.

Dataset	LLMs	Prompt Templates							
		t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8
WordNet	BERT-Large	2.19	9.36	9.18	19.41	4.72	19.34	9.93	27.85
	BART-Large	0.01	0.28	0.22	2.16	0.01	0.03	0.0	0.19
	Flan-T5-Large	0.17	19.70	5.54	31.26	0.0	3.03	5.70	26.80
	BLOOM-1b7	66.83	71.53	79.20	76.84	40.08	61.96	68.39	70.03
	Flan-T5-XL	2.81	40.26	17.83	52.21	0.01	7.75	18.47	18.85
	BLOOM-3b	63.33	75.29	79.08	77.06	37.40	65.32	68.99	71.62
	LLaMA-7B	37.26	74.61	70.16	75.97	24.28	76.61	67.41	81.38
	GPT-3	15.32	26.55	37.86	27.57	8.47	27.13	27.51	24.65
	GPT-3.5	24.27	80.81	89.46	91.72	0.81	60.76	49.38	82.41
	GPT-4	-	-	-	90.11	-	-	-	-
	<i>Flan-T5-Large</i> *	73.32	76.74	54.57	76.90	10.83	61.36	54.29	69.32
	<i>Flan-T5-XL</i> *	84.51	84.77	77.27	86.28	50.23	76.46	72.38	80.51
	BERT-Large	38.34	29.79	30.86	35.32	23.61	25.66	11.32	30.44
	BART-Large	8.47	0.57	2.23	0.98	21.48	20.51	7.83	23.21
GeoNames	Flan-T5-Large	11.55	3.57	13.16	4.68	9.45	6.05	8.17	7.38
	BLOOM-1b7	2.71	2.54	2.89	3.20	28.51	18.38	25.86	19.80
	Flan-T5-XL	33.81	15.71	19.77	20.78	15.36	12.41	18.43	15.82
	BLOOM-3b	3.76	4.70	2.64	3.43	28.84	18.08	25.64	20.71
	LLaMA-7B	29.49	14.16	25.54	15.95	13.91	9.44	17.79	16.79
	GPT-3	22.42	8.72	-	7.50	-	-	-	-
	GPT-3.5	35.00	-	-	-	-	-	-	-
	GPT-4	43.28	-	-	-	-	-	-	-
	<i>Flan-T5-Large</i> *	15.08	15.17	14.93	15.12	15.77	16.28	15.93	16.91
	<i>Flan-T5-XL</i> *	18.35	18.12	18.12	17.91	17.26	17.32	17.45	17.64
	BERT-Large	38.34	29.79	30.86	35.32	23.61	25.66	11.32	30.44
	BART-Large	8.47	0.57	2.23	0.98	21.48	20.51	7.83	23.21
	Flan-T5-Large	11.55	3.57	13.16	4.68	9.45	6.05	8.17	7.38
	BLOOM-1b7	2.71	2.54	2.89	3.20	28.51	18.38	25.86	19.80

表10. 零样本测试和微调的详细结果，涵盖七种用于本体学习任务B、类型分类发现的大语言模型，以F1分数表示。结果以百分比形式呈现。 * 表示微调模型的结果。

数据集 大语言模型		提示模板							
		t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8
GeoNames	BERT-Large	41.00	51.69	40.55	48.70	37.16	41.07	41.70	54.54
	BART-Large	38.11	41.03	40.55	52.50	39.09	45.80	36.67	55.40
	Flan-T5-Large	59.63	48.24	54.08	48.24	44.40	51.30	36.40	38.44
	BLOOM-1b7	33.16	31.04	33.16	32.83	33.77	33.53	36.67	32.92
	Flan-T5-XL	49.37	44.05	45.09	52.41	43.92	46.34	49.98	44.29
	BLOOM-3b	35.85	39.12	53.92	30.22	35.62	33.60	48.26	37.73
	LLaMA-7B	33.49	33.49	33.49	33.49	33.49	33.49	33.49	33.49
	GPT-3	43.43	51.74	42.70	53.20	46.04	52.56	45.49	52.62
	GPT-3.5	59.40	47.79	67.78	41.95	48.02	51.72	45.25	43.86
	GPT-4	38.56	52.46	34.00	38.89	44.06	55.43	33.78	36.23
	<i>Flan-T5-Large*</i>	42.53	59.40	40.29	62.46	46.03	57.41	42.49	62.04
	<i>Flan-T5-XL*</i>	48.41	34.80	55.23	46.96	57.48	36.29	59.05	49.26
UMLS	BERT-Large	48.21	38.84	41.46	40.41	45.88	40.91	41.04	42.92
	PubMedBERT	33.71	33.71	33.71	33.71	33.71	33.71	33.71	33.71
	BART-Large	36.02	48.21	41.42	49.90	39.37	47.47	42.39	45.46
	Flan-T5-Large	47.55	51.22	55.32	40.94	49.45	50.87	44.23	42.90
	BLOOM-1b7	33.71	36.18	33.71	38.26	33.71	35.89	33.27	33.60
	Flan-T5-XL	64.25	46.53	51.00	41.54	60.07	42.83	51.25	41.18
	BLOOM-3b	33.16	37.23	34.82	35.77	33.16	35.89	33.05	37.48
	LLaMA-7B	32.94	32.94	32.94	32.94	32.94	32.94	32.94	32.94
	GPT-3	51.58	49.41	49.86	42.90	50.57	46.07	45.36	46.72
	GPT-3.5	61.38	70.38	63.91	66.82	63.14	67.27	56.64	64.41
	GPT-4	41.19	76.99	42.55	63.88	50.28	78.11	36.59	60.72
	<i>Flan-T5-Large*</i>	37.17	48.66	36.07	42.12	48.39	46.65	53.42	35.97
	<i>Flan-T5-XL*</i>	63.69	50.04	36.91	41.34	78.12	50.12	79.25	39.27
Schema.Org	BERT-Large	43.85	41.17	44.06	43.20	43.70	40.05	42.15	43.72
	BART-Large	34.62	38.69	39.28	52.90	38.20	41.17	43.26	42.74
	Flan-T5-Large	46.98	49.92	46.11	54.78	40.27	54.47	42.06	47.93
	BLOOM-1b7	33.39	47.83	33.39	39.77	38.92	48.56	44.35	39.57
	Flan-T5-XL	42.70	33.45	33.59	42.76	36.69	34.04	33.75	36.45
	BLOOM-3b	41.64	47.16	47.98	45.25	39.73	40.75	51.28	48.73
	LLaMA-7B	33.37	33.37	33.37	33.37	33.37	33.37	33.37	33.37
	GPT-3	49.64	49.28	50.97	48.03	47.19	48.63	48.87	49.48
	GPT-3.5	56.84	74.38	58.52	70.16	53.35	72.35	54.16	71.03
	GPT-4	58.47	72.82	65.83	63.30	50.56	74.24	57.45	63.69
	<i>Flan-T5-Large*</i>	35.35	85.43	29.82	89.24	41.30	91.68	42.46	56.39
	<i>Flan-T5-XL*</i>	91.06	57.46	74.68	65.32	91.54	50.63	91.70	33.33

Table 10. The detailed results of zero-shot testing and finetuning across seven LLMs reported for ontology learning Task B, type taxonomy discovery in F1-score. The results are in percentages. The * denotes finetuning model results.

Dataset	LLMs	Prompt Templates							
		t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8
GeoNames	BERT-Large	41.00	51.69	40.55	48.70	37.16	41.07	41.70	54.54
	BART-Large	38.11	41.03	40.55	52.50	39.09	45.80	36.67	55.40
	Flan-T5-Large	59.63	48.24	54.08	48.24	44.40	51.30	36.40	38.44
	BLOOM-1b7	33.16	31.04	33.16	32.83	33.77	33.53	36.67	32.92
	Flan-T5-XL	49.37	44.05	45.09	52.41	43.92	46.34	49.98	44.29
	BLOOM-3b	35.85	39.12	53.92	30.22	35.62	33.60	48.26	37.73
	LLaMA-7B	33.49	33.49	33.49	33.49	33.49	33.49	33.49	33.49
	GPT-3	43.43	51.74	42.70	53.20	46.04	52.56	45.49	52.62
	GPT-3.5	59.40	47.79	67.78	41.95	48.02	51.72	45.25	43.86
	GPT-4	38.56	52.46	34.00	38.89	44.06	55.43	33.78	36.23
	<i>Flan-T5-Large*</i>	42.53	59.40	40.29	62.46	46.03	57.41	42.49	62.04
	<i>Flan-T5-XL*</i>	48.41	34.80	55.23	46.96	57.48	36.29	59.05	49.26
UMLS	BERT-Large	48.21	38.84	41.46	40.41	45.88	40.91	41.04	42.92
	PubMedBERT	33.71	33.71	33.71	33.71	33.71	33.71	33.71	33.71
	BART-Large	36.02	48.21	41.42	49.90	39.37	47.47	42.39	45.46
	Flan-T5-Large	47.55	51.22	55.32	40.94	49.45	50.87	44.23	42.90
	BLOOM-1b7	33.71	36.18	33.71	38.26	33.71	35.89	33.27	33.60
	Flan-T5-XL	64.25	46.53	51.00	41.54	60.07	42.83	51.25	41.18
	BLOOM-3b	33.16	37.23	34.82	35.77	33.16	35.89	33.05	37.48
	LLaMA-7B	32.94	32.94	32.94	32.94	32.94	32.94	32.94	32.94
	GPT-3	51.58	49.41	49.86	42.90	50.57	46.07	45.36	46.72
	GPT-3.5	61.38	70.38	63.91	66.82	63.14	67.27	56.64	64.41
	GPT-4	41.19	76.99	42.55	63.88	50.28	78.11	36.59	60.72
	<i>Flan-T5-Large*</i>	37.17	48.66	36.07	42.12	48.39	46.65	53.42	35.97
	<i>Flan-T5-XL*</i>	63.69	50.04	36.91	41.34	78.12	50.12	79.25	39.27
schema.org	BERT-Large	43.85	41.17	44.06	43.20	43.70	40.05	42.15	43.72
	BART-Large	34.62	38.69	39.28	52.90	38.20	41.17	43.26	42.74
	Flan-T5-Large	46.98	49.92	46.11	54.78	40.27	54.47	42.06	47.93
	BLOOM-1b7	33.39	47.83	33.39	39.77	38.92	48.56	44.35	39.57
	Flan-T5-XL	42.70	33.45	33.59	42.76	36.69	34.04	33.75	36.45
	BLOOM-3b	41.64	47.16	47.98	45.25	39.73	40.75	51.28	48.73
	LLaMA-7B	33.37	33.37	33.37	33.37	33.37	33.37	33.37	33.37
	GPT-3	49.64	49.28	50.97	48.03	47.19	48.63	48.87	49.48
	GPT-3.5	56.84	74.38	58.52	70.16	53.35	72.35	54.16	71.03
	GPT-4	58.47	72.82	65.83	63.30	50.56	74.24	57.45	63.69
	<i>Flan-T5-Large*</i>	35.35	85.43	29.82	89.24	41.30	91.68	42.46	56.39
	<i>Flan-T5-XL*</i>	91.06	57.46	74.68	65.32	91.54	50.63	91.70	33.33